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Abstract

Skin diseases remain a global health challenge affecting billions worldwide. Due to the prevalence of skin diseases, automated diagnostic systems have become increasingly adopted. However, these systems often underperform on darker skin tones and visually similar conditions. Moreover, many existing studies prioritise accuracy on curated datasets, while overlooking balanced evaluation metrics, minority class performance, and fairness. This limits the reliability of diagnostic systems across diverse patient groups and highlights the need for equitable classification approaches.

This study proposes a hybrid deep learning framework that integrates ConvNeXt-Small and Swin-Small. An Attentional Feature Fusion (AFF) mechanism combines local convolutional features with global attention features to detect ten skin diseases from the Kaggle dataset (Hossain, 2021). Furthermore, a ResNet-50 baseline is applied under identical conditions for comparison.

The proposed model achieved 90.27% accuracy and a macro F1-score of 0.866, illustrating an improvement of 11.79 percentage points (pp) compared to the baseline. Most importantly, recall improved for classes including Psoriasis/Lichen Planus and Melanocytic Nevi, while skin tone bias was reduced from approximately 8.0pp to 2.68 pp, demonstrating that improvements extended beyond accuracy results. Moreover, the study introduced an evaluation framework encompassing fairness evaluation, calibration analysis, and class-wise robustness assessment, highlighting the importance of moving beyond accuracy-focused evaluation when developing reliable AI systems.

Keywords: skin disease classification, hybrid deep learning, ConvNeXt, Swin Transformer, ResNet, attentional feature fusion, fairness, skin tone, dermoscopy, confidence calibration

Contents

Abstract	1
List of Figures	4
1 Introduction	7
1.1 Background	7
1.2 Challenges in Dermatological Diagnosis	7
1.3 Overview of Proposed Approach	8
1.4 Aim	8
1.5 Research Objectives	9
2 Literature Review	9
2.1 Image Classification Research in Medical Imaging	9
2.2 Traditional Image Classification Approaches	9
2.3 Deep Learning Approaches	10
2.4 Transformer-Based Models	13
2.5 Hybrid Models	14
2.6 Evaluation and Benchmarking	15
2.7 Data Augmentation Techniques	17
2.8 Research Gap and Motivation	18
3 Methodology	19
3.1 Methodological Framework	19
3.2 Dataset and Data Preparation	19
3.3 Model Architecture	23
3.4 Training Strategy	28
3.5 Evaluation Framework	31
3.6 Implementation Details	33

3.7	Project Management	36
4	Results	38
4.1	Overall Classification Performance	38
4.2	Baseline Model Performance (ResNet-50).....	40
4.3	Per-Class Performance Analysis	43
4.4	Attentional Feature Fusion Analysis.....	44
4.5	Confidence Calibration Analysis.....	45
4.6	Skin Tone Fairness Analysis	46
4.7	Grad-CAM Explainability Analysis	47
4.8	Comparative Analysis with Published Literature	48
4.9	Evaluation Limitations.....	49
5	Discussion.....	50
5.1	Architectural Contributions to Performance	50
5.2	AUROC–Macro F1 Trade-off Evaluation	51
5.3	Fairness and Bias Mitigation	52
5.4	Clinical Trustworthiness: Calibration and Explainability	53
5.5	Limitations and Future Work	54
6	Conclusion.....	56
6.1	Revisiting Aims and Objectives	56
6.2	Evaluation of Research Objectives	56
6.3	Contribution to the Research Field	57
6.4	Final Reflections and Future Directions	57
6.5	Concluding Remarks.....	58
	References.....	59
	Appendices.....	64

List of Figures

Figure 3.1 Visual example999s of skin disease classes from dataset	20
Figure 3.2 Bar graph and pie chart displaying class distribution and proportion respectively	21
Figure 3.3 Visualisation of TTA-enabled transformations such as rotation	22
Figure 3.4 Hybrid ConvNeXt+Swin-Small architecture with AFF gate.....	25
Figure 3.5 ResNet-50 Architecture with bottleneck detail.....	26
Figure 3.6 Adjustment of dropout rate from 0.4 to 0.3 during implementation refinement..	29
Figure 3.7 Summary of the training outcomes	29
Figure 3.8 Screenshot of the total number of training cycles for ResNet-50 baseline	30
Figure 3.9 Class diagram of the proposed hybrid model architecture, illustrating the relationships between the hybrid network components, feature fusion mechanism, and supporting modules	30
Figure 3.10 Examples of misclassified skin lesions alongside their Grad-CAM visualisations, which highlights the regions the model focused on when making predictions	33
Figure 3.11 Digital sketch of the Skin Disease Classifier interface, showing image upload, classification results, confidence scores, and feedback functionality for misclassified images	34
Figure 3.12 Gantt Chart illustrating the project timeline (over a 12-week period) for skin disease classification system.....	37
Figure 4.1 Per-Class comparison bar charts between hybrid and baseline model across the metrics recall, precision, and F1 Score.....	40
Figure 4.2 ResNet-50 Baseline Confusion Matrix.....	42
Figure 4.3 ResNet-50 Per-Class performance across recall, precision, and F1 Score	42
Figure 4.4 ConvNeXt-Swin per-class binary confusion matrices.....	44
Figure 4.5 ResNet-50 per-class binary confusion matrices	44
Figure 4.6 Visualisation of AFF gate analysis.....	45
Figure 4.7 Confidence calibration output	46
Figure 4.8 Visualisation of Grad-CAM on lesion	48
Figure B1 Hybrid ConvNeXt-Swin Confusion Matrix	Error! Bookmark not defined.
Figure C1 ConvNeXt-Swin Hybrid Training Curves.....	53
Figure C2 ResNet-50 Baseline Training Curves.....	53

Figure D1 Per-Class Precision, Recall, and F1-Score (ConvNeXt–Swin Hybrid Model).....	54
Figure D2 Skin Tone Fairness Analysis (Accuracy and Macro F1 by ITA Group).....	54
Figure D3 Skin Tone Representation Across Selected Classes	54
Figure D4 ResNet-50 Accuracy by Estimated Skin Tone (ITA).....	55

List of Tables

Table 3.1 Skin Disease Classes and Image Counts	19
Table 3.2 Experimental Conditions	26
Table 3.3 Shared Training Hyperparameters.....	27
Table 3.4 Functional and Non-Functional Requirements.....	35
Table 3.5 User Story Outline	36
Table 4.1 Overall performance metrics.....	39
Table 4.2 Per-class recall comparison	43
Table 4.3 Literature comparison	48
Table A1 Evaluation Metrics Glossary.....	51

1 Introduction

1.1 Background

Skin diseases represent a significant global health burden that affects a substantial proportion of the world's population. According to The World Health Organisation, 'over 3,000 skin diseases affect between 1.8 billion and 4.8 billion people' (Trakatelli *et al.*, 2023). Although genetic predisposition contributes to susceptibility, environmental and lifestyle factors also play a significant role in the development of skin conditions (Cirstea *et al.*, 2025). In addition, rising healthcare costs increase pressure on healthcare systems (Bajwa *et al.*, 2020), yet skin health remains 'disproportionately disregarded in international health strategies' (Ni *et al.*, 2026).

1.2 Challenges in Dermatological Diagnosis

Skin diseases are frequently chronic which can increase the risk of impacted individuals suffering from anxiety and depression. As a result of this reduced quality of life, early detection is crucial for lessening the burden at the individual and institutional level. In severe cases such as melanoma, failure to accurately diagnose the disease on time can be life-threatening. This is particularly concerning given the estimate of 132,000 cases of melanoma annually (Urban *et al.*, 2021).

Despite widespread recognition of the need for early detection, implementing this into practice remains a significant challenge. In the United Kingdom, limitations in National Health Service's capacity have prolonged waiting times and increased diagnostic backlogs. In some cases, in-person consultations are delayed by several months. Additional delays can arise from diagnostic variability between general practitioners and dermatologists, which further exacerbates clinical experiences.

These healthcare challenges have driven the growing implementation of technology-based diagnostic tools. Among these tools, machine learning models are increasingly implemented to support clinical decision-making; however, their reliability remains under discussion. Despite technological advancements, disease misclassification remains a significant challenge, particularly among darker skin tone populations. This is largely due to training

dermatological datasets often containing disproportionately larger amounts of samples for light skin populations. Therefore, while classification models can exhibit strong accuracy performance, there remains a higher chance of classification errors when applied to dark-skinned patients.

Benčević highlights that many researchers must often rely on indirect measures for skin tone, such as the Fitzpatrick scale or individual typology angle (ITA) (Benčević *et al.*, 2024). However, this approach is often neglected in the research field as previous studies have yet to adequately address representation bias and typically focus solely on enhancing the performance benchmark of a model. Furthermore, additional challenges in heterogeneous datasets, such as class imbalance and the visual similarity between different lesion types, further complicate model performance. These limitations reduce confidence in model generalisability across clinical populations.

1.3 Overview of Proposed Approach

Deep learning approaches offer a potential solution to address the limitations associated with conventional diagnostic methods. Classifying diverse skin lesion types remains challenging due to complex visual features including colour, shape, and texture. Providing reliable diagnosis across diverse skin tones requires both an effective architecture and a broader evaluation strategy to account for variation across patient groups.

Consequently, the dissertation proposed a deep learning framework that distinguishes between different types of dermatological diseases including skin cancer across diverse skin pigmentations. Furthermore, to ensure a fair assessment of the model's performance, the training and testing pipeline incorporated relevant grouping and labelling techniques to address class imbalance and representation bias. In doing so, the evaluation determined the extent to which the model can be generalised and applied in real-world situations as a clinical assistance tool. Moreover, the introduced diagnostic tool could help healthcare facilities such as the NHS in carrying out automated medical triage (Halawani *et al.*, 2025).

1.4 Aim

This research aims to build a deep learning framework that distinguishes between different types of dermatological diseases including skin cancer across diverse skin pigmentations.

1.5 Research Objectives

- To examine published research on traditional machine learning models, deep learning architectures, and hybrid models for medical image classification.
- To build a hybrid architecture capable of capturing both local and broader image representations.
- To develop a robust evaluation framework that incorporates accuracy, macro-averaged metrics, area under the receiver operating characteristic curve (AUROC), and calibration measures.
- To compare the proposed model with a baseline under identical experimental conditions.
- To develop a prototype web application demonstrating deployment of the trained hybrid model.
- To evaluate methodological limitations and identify directions for future research.

2 Literature Review

2.1 Image Classification Research in Medical Imaging

This review examines automated image classification methods ranging from conventional machine learning to contemporary architectures that leverage deep learning algorithms. Attention is allocated to models that are employed in skin disease detection, particularly in tasks involving visually similar lesions, the handling of limited labelled data, and the consideration of skin tone variation. In addition, it considers the implementation of data augmentation and different evaluation approaches to improve model reliability and support their applicability in real clinical settings.

2.2 Traditional Image Classification Approaches

Traditional machine learning approaches have historically been applied to medical imaging, where diagnosis is visually complex. These conventional models rely on handcrafted feature (i.e. edges, shapes, or textures) extraction techniques from raw data. A few common methods include Histogram of Oriented Gradients (HOG), and Bag-of-Visual-Words (BoVW)

(Alhindi *et al.*, 2018), which are subsequently classified by models such as Support Vector Machines (SVM), K-Nearest Neighbours (KNN), and decision trees (DT).

Among these methods, SVM has been known to be an efficient one in high-dimensional spaces due to its efficiency in maximising class separation (Chandra and Bedi, 2021). Alternatively, random forests improve the predictive stability by combining many decision trees trained on random subsets of the data. However, empirical evidence illustrates that performance differences between classifiers are often marginal when feature representations are fixed. Lorente, for example, reported almost identical results between k-NN (0.74), logistic regression (0.79), and SVM (0.83) using Bag-of-Visual-Words attributes, with SVM performing only modestly better (Lorente, Riera and Rana, 2021). This establishes that specific data preprocessing steps as well as feature representation may have a greater impact than the choice of the classifier.

Despite their early success, such conventional approaches are fundamentally limited due to their dependency on fixed handcrafted techniques. As they cannot adapt throughout the training process, their ability to recognise context-dependent patterns is restricted, particularly in heterogeneous clinical datasets. Consequently, these models exhibit limited robustness under varying conditions. Performance may decline across devices, lighting conditions, and skin tones. Therefore, traditional models remain useful as baseline models, however, are insufficient for handling the complexities associated with contemporary classification tasks.

2.3 Deep Learning Approaches

Deep learning approaches enable models to learn feature representations directly from the data which eliminates the requirement for manual feature engineering. This technique has consistently demonstrated strong performance in image classification tasks, where defining relevant features is inherently challenging. On the contrary, early fully connected neural networks were not well-suited for image data, as they cannot effectively learn the spatial relationships between pixels such as the arrangement of edges or texture features. Consequently, many parameters are required which makes traditional models computationally inefficient and prone to overfitting.

Convolutional Neural Networks

Convolutional Neural Networks (CNNs) were introduced to address the shortcomings of conventional classifiers. By leveraging convolutional filters, shared weights, and localised feature extraction, CNN models can preserve spatial information within images. This structure enables the model to learn features in a layered (hierarchical) manner, allowing meaningful simple features such as edges to be captured by lower layers, while complex features are captured by deeper layers (Lecun *et al.*, 1998).

The effectiveness of CNNs in image classification tasks has been demonstrated by several models such as AlexNet in 2012, which performed significantly strong and achieved an error rate of 15.3% (Krizhevsky, Sutskever and Hinton, 2017). Further refinements have since been made by subsequent models such as Visual Geometry Group (VGG) networks which have shown that increasing depth with smaller (3x3) convolutional filters produce greater accuracy. On the contrary, ResNet has incorporated skip connections that facilitate learning in deep networks without experiencing a decline in model performance over time (Liu, 2023). Despite CNN models improving benchmark performance in image classification, their effectiveness is influenced by additional factors such as architectural design and dataset characteristics which are explored further in subsequent sections.

ResNet

ResNet has become a widely adopted model in medical image classification due to its ability to train deep networks effectively by leveraging residual (skip) connections (Xu, Fu and Zhu, 2023). These skip connections behave as 'shortcuts' and allow information to be transferred by skipping layers and reducing the rate in which a model's performance declines as the networks grow deeper. As a result, this architecture enables ResNet to learn more detailed and complex features, which is critical when identifying subtle visual patterns.

For example, Castilla reported a high AUC score of 0.955 by applying ResNet18 for diabetic retinopathy detection, which demonstrated its proficiency in distinguishing among different classes (Castilla *et al.*, 2022). Similarly, ResNet-50, containing 50 layers, is frequently selected as a baseline model due to its 'balance between computational efficiency and its capacity to learn details' (Najla Mohamed Salh Kailani, 2025). In addition, its residual connections ensure consistent performance across a range of classification tasks (Borawar

and Kaur, 2023). Nevertheless, strong performance in controlled settings cannot guarantee clinical reliability as critical factors including class imbalance and additional relevant evaluation metrics, are often overlooked.

EfficientNet

EfficientNet takes a fundamentally different approach by adopting a compound scaling strategy that balances network depth, width, and input resolution using fixed scaling coefficients. This architecture enables a strong performance with relatively fewer parameters which is particularly useful for transfer learning in medical imaging, where labelled data is limited (Tan and Le, 2020).

In dermatological applications, EfficientNet's scaling method ensures that the lesion details are maintained at a higher resolution which helps preserve fine-grained lesion details required for distinguishing subtle visual differences. However, despite its efficiency, the EfficientNet network exhibits limitations under several conditions. For instance, EfficientNet models are often validated on a single train-validation split, which may not be reliable in datasets having duplicate backgrounds or inconsistent label values. Furthermore, despite being 'efficient' in relation to its parameter count, such models can be slow to train and run, and certain variants require a large amount of GPU memory. Consequently, although EfficientNet offers a strong balance between accuracy and efficiency, its practical reliability depends on careful experimental design and robust evaluation.

ConvNeXt

ConvNeXt was introduced as a modernised convolutional approach that combined several design components from Vision Transformers, while retaining the inductive biases of CNNs. This was accomplished by applying larger convolutional filters, inverted residual blocks, and additional techniques to capture and retain spatial patterns.

Recent studies highlight the efficiency of ConvNeXt in classifying dermatological images by capturing texture, edge, and colour variations within skin lesions (Aruk, Pacal and Toprak, 2025). For example, Özdemir and Pacal reported 93.48% accuracy, 93.24% precision, and 90.70% recall in skin lesion classification (Ozdemir and Pacal, 2025). Additionally, Mmileng's research involving classifying malaria parasite images reported an accuracy rate above 98%,

higher than that obtained by ResNet-50 and VGG16 (Mmileng *et al.*, 2025). Notwithstanding ConvNeXt's consistency, the architecture remains less researched in comparison to other approaches, such as ResNet and EfficientNet. This suggests a requirement for further study to examine the robustness of ConvNeXt under different experimental conditions.

These developments reflect a shift in image classification from strictly convolutional methods to methods that incorporate elements of global contextual information. However, given that ConvNeXt and other CNNs remain limited in their ability to model long-range dependencies, this has driven the growing adoption of transformer-based architectures.

2.4 Transformer-Based Models

Transformer-based models have emerged in response to the limitations of convolutional networks, particularly in their restricted ability to model non-local relationships within the data. By leveraging self-attention mechanisms, transformers can model global interactions across an image, which allows relationships between distant regions to be learned more effectively. In dermatological applications, where features such as lesion asymmetry are clinically significant, this capability allows models to analyse spatial patterns more holistically.

In contrast to CNNs, which rely on localised feature extractions, transformers process images as a series of patches and calculate attention across the entire input. Among this model family, Vision Transformers (ViT) and Swin Transformers have become prominent architectures in medical imaging.

Vision Transformer Models

Vision Transformers (ViT) extend this approach by applying self-attention across all image patches, which facilitates a global understanding of the image beyond the local biases of CNNs (Dosovitskiy *et al.*, 2021). While this has led to 'state-of-the-art' performances in large-scale classification tasks, their weak inductive biases can also present challenges including lack of local feature bias, sensitivity to hyperparameters, and longer training times compared to CNNs. Additionally, ViTs often require large amounts of data or pretraining to surpass the performance of CNNs that have been carefully optimised. This limits their applicability in medical imaging settings where labelled data is difficult and expensive to

obtain (Khalil, Khalil and Ngom, 2023). Consequently, despite their strengths, the practical use of Vision Transformers remains constrained by data and training requirements.

Swin Transformer Models

The Swin Transformer improves upon standard transformer architectures by introducing hierarchical feature representation and a shifted window mechanism for self-attention. Contrary to Vision Transformers, which apply global attention across all patches, Swin Transformers constrain attention within local windows while allowing connections across windows. Resultantly, this design reduces computational complexity and improves scalability, which makes it more suitable for processing high-resolution images (Liu *et al.*, 2021).

In relation to skin disease detection, Pacal reported strong performance in skin cancer detection (i.e. accuracy of 89.36%, a recall of 85.13%, a precision of 88.22%, and an F1 score of 86.65%) (Pacal, Alaftekin and Zengul, 2024). These findings highlight the effectiveness of hierarchical attention mechanisms in modelling subtle visual patterns. Nonetheless, despite such improvements, Swin Transformers involve a trade-off between efficiency and global context modelling. Although the shifting window enables indirect communication between image regions, it remains less efficient than full self-attention. Additionally, the computational demands and large data requirements of such models present practical challenges in medical imaging, where data availability is often limited. Taken together, research confirms that despite the remarkable feature representation capability of transformers, their disadvantages necessitate a hybrid approach where the efficiency of CNNs is combined with the global contextualisation capability of transformers.

2.5 Hybrid Models

In recent years, hybrid architectures have emerged to combine the complementary strengths of convolutional and transformer-based models. These approaches aim to leverage the strong local feature extraction and inductive bias of CNNs alongside the global contextual modelling of transformers. Several studies have demonstrated the effectiveness of such approaches in medical imaging. For instance, hybrid models such as DenseNet-ViT, have shown strong performance with medical imaging datasets as seen with Chappidi and Srinivas who reported a high accuracy score of 96.02% (Chappidi and Srinivas, 2025). This

integration has also been explored with modern convolutional architectures including ConvNeXt. However, the ConvNeXt paradigm does not utilise the full potential of transformers and is mainly operating within its local receptive field.

The combination of ConvNeXt and Swin Transformer is therefore methodologically justified for skin disease detection as both architectures offer complementary strengths. ConvNeXt effectively captures details such as pigmentation and texture, while Swin Transformer captures asymmetrical lesions and their relationship with other areas of the skin. Both theoretical and empirical evidence suggest that hybrid architectures can improve model robustness across different imaging conditions, often outperforming monolithic models. Notwithstanding, no study to date has evaluated ConvNeXt–Swin hybrid while simultaneously addressing class imbalance, skin tone stratification, and calibration analysis.

Kılıç, for instance, applied a dual-path ConvNeXt–Swin model for blood cell classification and reported near perfect classification accuracy (i.e. 99.98%) (Kılıç, 2025). The study also revealed strong performance across multiple evaluation metrics, demonstrating robust discrimination between visually similar classes through joint local and global feature modelling. Having said that, it is important to note that such a high accuracy can be expected from blood cell classification since it is a substantially easier task than ten-class skin disease classification, for example, due to more distinct morphological differences. Additionally, a critique of the methodological approach can be likened to the lack of explicit additional metrics, such as precision, recall, F1-score, confusion matrices, and class-wise performance, which restricts insight into model behaviour across classes and demographic groups. Therefore, this research must be interpreted with caution when comparing its performance to more complex medical image classification tasks.

2.6 Evaluation and Benchmarking

Despite significant advances in model architectures, the evaluation of dermatological classification systems remains a critical challenge. Model performance is highly dependent on both dataset characteristics and the choice of evaluation metrics. Furthermore, skin disease datasets are challenging to analyse due to their imbalanced, visually heterogeneous, and demographically unrepresentative state. Consequently, models may achieve high overall

accuracy by prioritising majority classes, while underperforming on clinically important minority conditions.

This limitation is particularly evident in datasets such as the Kaggle Skin Diseases Image Dataset (Hossain, 2021), where class imbalance undermines the significance of accuracy as an individual measure. In this regard, a more informative evaluation can be achieved through the adoption of class-wise recall, macro F1 score (i.e. measurement of how well a model balances precision and recall across all classes equally), sensitivity, specificity, and AUC. This is particularly critical in cases where diverse skin tone samples are present.

Several studies have developed deep learning models to detect lesion images belonging to the Kaggle Skin Diseases Image Dataset that contains 27,153 images and ten classes (Hossain, 2021). Alruwaili and Mohamed, for example, proposed a multiclass classification approach which consisted of EfficientNet-B0, EfficientNet-B2, and ResNet-50 (Alruwaili and Mohamed, 2025). The researchers achieved a high accuracy of 99.14% based on an 80-10-10 split. However, the study relied on a single test split without cross-validation or external testing, which may limit confidence in the model's generalisability. In addition, the study overlooked performance across different skin tones, which brings into question the model's fairness and demographic robustness. Likewise, Fernández-Gutiérrez achieved strong performance using ResNet-152, with training and testing accuracies of 95% and 92.3%, respectively (Fernández-Gutiérrez *et al.*, 2025). Considering the modest performance gap, this indicates strong generalisability. Having said that, the absence of cross validation or external testing makes it unclear how well the model would perform on truly unseen data.

Additionally, preprocessing techniques introduce yet another challenge to performance evaluation. As highlighted by Noor, adjusting contrast and brightness can improve image quality, and, in turn, boost model performance. These findings are supported by the increase in peak signal-to-noise ratio (PSNR) and reduced mean squared error (MSE), which allows the model to detect features sufficiently (Noor *et al.*, 2025). However, if these adjustments are overused, they may introduce artificial patterns that may distort the true appearance of skin lesions.

These studies reveal both the strengths and limitations of current approaches applied to the Kaggle dataset. As discussed, many studies rely on single train-test splits, provide limited

analysis of minority classes and skin tone variation, and heavily depend on preprocessing techniques. As a result, strong benchmark performance may not reflect real clinical outcomes. Although some studies report higher accuracy values, evaluation protocols often differ substantially across studies. For instance, calibration analysis is often excluded by such studies, despite its significance during the evaluation process as it ‘ensures that a model’s estimated probabilities match real-world likelihoods’ (Pavlovic, 2025). Therefore, broader evaluation strategies are required to assess model reliability beyond accuracy alone.

2.7 Data Augmentation Techniques

Beyond architectural design, improving model performance in skin disease classification also depends heavily on the quality and diversity of the training data. Data augmentation techniques are, therefore, essential in addressing limited training data and class imbalance. Methods such as rotation, flipping, and cropping can increase data diversity without distorting the underlying labels (Kumar *et al.*, 2025). As observed by Noor, in dermatology, colour and lighting adjustments are commonly employed (Noor *et al.*, 2025). Pacal stressed the importance of incorporating data augmentation methods to ensure the relevant feature retention of lesions (Pacal, Alaftekin and Zengul, 2024). However, there are certain methodological risks involved in the process of data augmentation such as the possibility of losing significant information concerning pigmentation and erythema with significant colour change.

Advanced techniques such as generative adversarial networks (GANs) have been increasingly implemented to improve representation of minority classes. However, there are many technical and ethical issues associated with such approaches. For instance, there is a possibility that the output data might contain artifacts, leading to increased bias in the current data or misrepresentation of the data itself. Therefore, while the addition of more data via augmentation could potentially assist with data diversification, this may not always be the case.

Improving dermatological AI systems requires moving beyond data quantity and benchmark accuracy. More critical considerations include generalisation, fairness, and interpretability, as well as minority class performance under real world conditions. This is particularly critical given that many studies still do not go beyond single split evaluation and fail to report

demographic distributions, especially with respect to skin tone. Therefore, architectural performance alone is insufficient to establish clinical reliability.

2.8 Research Gap and Motivation

This review presents several critical limitations in contemporary dermatological image classification research. Although convolutional, transformer, and hybrid models have improved classification performance, many studies still rely on limited evaluation methods and biased datasets. As a result, it is difficult to establish the accuracy and reliability of the models as their efficiency may not be reliable for real-world applications.

A key limitation lies in the absence of comprehensive evaluation criteria that explicitly accounts for class imbalance, minority class performance, calibration, and demographic variation, particularly across different skin tones. As a result, this contributes to an inadequate assessment of fairness and a lack of calibration analysis. Therefore, the absence of this measurement confines both interpretability and clinical trust. Furthermore, despite hybrid models demonstrating strong potential, their effectiveness remains insufficiently explored under unbiased conditions, compared to comparative models. Therefore, additional research is required.

To address these limitations, this study proposes a hybrid deep learning framework that combines convolutional and transformer networks for feature extraction alongside a broader evaluation strategy. In addition to conventional performance metrics, the evaluation incorporates calibration, fairness analysis, interpretability, and class-wise assessment across skin tone groups. Although this study does not fully resolve all generalisation challenges in dermatological AI, it facilitates the expansion of evaluation depth through calibration, fairness, interpretability, and controlled baseline comparison. By combining model design with a broader evaluation strategy, this research aims to evaluate whether high-performing models can also maintain reliability under clinically relevant conditions.

3 Methodology

3.1 Methodological Framework

The methodology structure consisted of four main components, including dataset and data preparation, model architecture, training strategy, and evaluation framework. The proposed architecture combined ConvNeXt and Swin Transformer backbones. ConvNeXt captured localised features, while Swin Transformer modelled global context. The hybrid architecture was then compared to ResNet-50 baseline.

Although cross-validation and external testing can strengthen confidence in a model's ability to generalise, practical limitations such as time and computational resources often require studies to rely on a fixed train-validation-test split instead. Consequently, this study focused on extending evaluation landscape through calibration, fairness, and interpretability analysis under controlled experimental conditions.

3.2 Dataset and Data Preparation

The dataset employed was the Skin Diseases Image Dataset sourced from Kaggle (Hossain, 2021), containing 27,153 images across ten distinct disease classes, as outlined in Table 1. In contrast to existing studies that relied on curated and balanced datasets, this dataset was selected for its inherent class imbalance, offering a more realistic reflection of clinical conditions. As a result of the dataset's raw characteristics, this enabled a stronger evaluation of minority-class performance and model generalisation. Figure 3.1 presents representative samples from selected classes, illustrating the visual diversity and complexity of the dataset.

Table 3.1 Skin Disease Classes and Image Counts

Skin Disease Class	Image Count
Eczema	1,677
Warts, Molluscum and other Viral Infections	2,103
Melanoma	3,140

Skin Disease Class	Image Count
Atopic Dermatitis	1,257
Basal Cell Carcinoma (BCC)	3,323
Melanocytic Nevi (NV)	7,970
Benign Keratosis-like Lesions (BKL)	2,079
Psoriasis, Lichen Planus and related diseases	2,055
Seborrheic Keratoses and other Benign Tumors	1,847
Tinea, Ringworm, Candidiasis and other Fungal Infections	1,702



Figure 3.1 Visual examples of skin disease classes from dataset

Class Imbalance Challenges

One of the key limitations associated with this dataset was the substantial imbalance between classes. For instance, Melanocytic Nevus (NV) contained 7,970 images, while Atopic Dermatitis contained 1,257 images; hence the imbalance was at a ratio of about 6.3 to 1. Although this distribution mirrors real-world prevalence, it posed the risk of introducing bias during training, which could have caused the model to favour the majority classes. To mitigate this issue, appropriate data handling and training strategies were applied to support more balanced learning as well as ensure that performance was assessed more fairly across all classes.

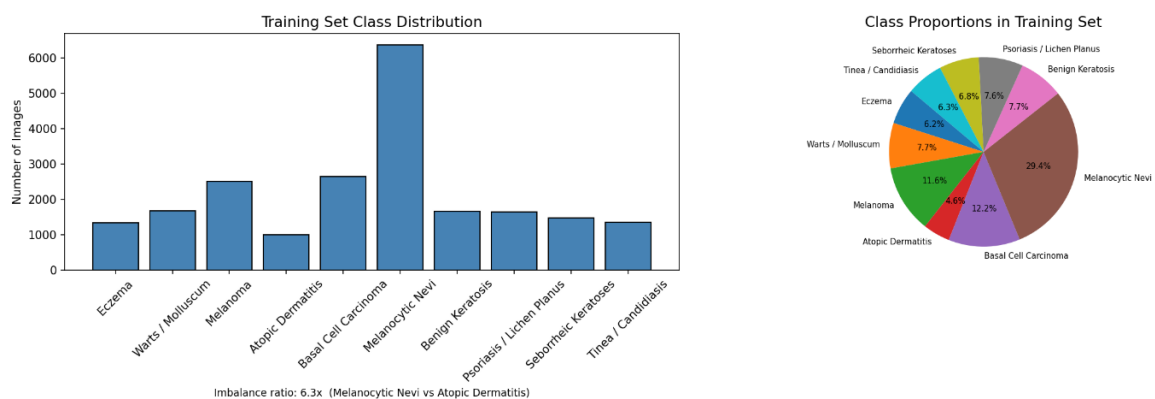


Figure 3.2 Bar graph and pie chart displaying class distribution and proportion respectively

Skin Tone Limitation

A further challenge could be attributed to the absence of skin colour labels, which were essential for assessing model fairness across demographic groups. Markedly, this drawback is widely observed among public dermatology datasets (Benčević *et al.*, 2024), where an absence of metadata prohibits researchers from identifying potential disparities in diagnostic outcomes across different skin tones. To address this limitation, the evaluation framework employed a proxy-based method to estimate skin tone for evaluation purposes. It is important to note that while this method did not replace clinically verified labels, it enabled a structured analysis of model behaviour across inferred skin tone groups, which facilitated a broader understanding of the model's behaviour.

Data Splitting

The dataset was divided into training, validation, and test sets using an 80/10/10 split. To ensure reproducibility, a fixed random seed (seed=42) was applied across Python modules

NumPy, and PyTorch. This three-way split separated model training, hyperparameter optimisation, and final evaluation, which reduced the risk of data leakage.

During training, early stopping was implemented at the validation phase, which is a technique that ‘stops model training when overfitting signs appear’ (Hussein and Shareef, 2024). This would allow training to be terminated if the loss calculated on the validation set did not improve by at least 0.0005 over fifteen consecutive epochs (i.e. training cycle). Additionally, a checkpoint strategy that preserves intermediate models was employed to retain the top three model checkpoints based on validation accuracy, ensuring that the most stable and well-performing models were preserved.

During the final evaluation stage, predictions were obtained by aggregating the outputs from several checkpointed models and applying test-time augmentation (TTA). TTA enhanced accuracy by randomly modifying test images (e.g. flipping, rotations), followed by passing them through the model. The trained model produced predictions on several variations of a test image and averaged the outputs to reduce noise sensitivity, while improving stability and reliability of the results (Kimura, 2021).



Figure 3.3 Visualisation of TTA-enabled transformations such as rotation

3.3 Model Architecture

Hybrid ConvNeXt-Swin Model

The proposed model adopted a dual-path hybrid architecture that combined ConvNeXt-Small and Swin Transformer-Small backbones to capture complementary features. ConvNeXt-Small primarily captured local visual patterns including texture, colour differences, and lesion boundaries (Liu, 2023), while Swin Transformer-Small modelled broader spatial relationships and structural context across an image. This fusion enabled the model to consider how different regions of a lesion relate to one another, which contributed to improved discrimination among commonly confused class pairs that had similar local textures but distinct morphology (e.g. Melanoma, Basal Cell Carcinoma, and Melanocytic Nevi). In addition, Swin’s shifted window attention enabled the model to recognise how different segments of the image related to each other (Liu *et al.*, 2021).

To effectively combine these representations, an Attentional Feature Fusion (AFF) mechanism was employed. The two feature vectors were concatenated into a single 1536-dimensional representation, which was then passed through two layers of a gating neural network. The first layer reduces the number of dimensions (i.e. features) of the input data from 1536 to 768. This was followed by a Rectified Linear Unit (ReLU) activation to ensure that the output was always positive and introduced non-linearity. In the second layer, a Sigmoid (σ) activation function was applied to generate gating weights in the range [0,1], which enabled the model to adaptively control the contribution of each feature representation (Dai *et al.*, 2021). This advantage was the motivation behind selecting the AFF gate for this specific task in detecting visually similar skin diseases.

This stage produced a weighting vector, α , that assigned an importance score to each feature (Dai *et al.*, 2021). By relying on this information, adjustments were made to the contribution of each feature set, which enabled the model to concentrate on the most relevant local and global aspects for a specific sample input. This flexibility allowed the model to understand different images based on its adjustments for each image individually. Consequently, this improved its capacity to differentiate across illness classifications that might have identical visual characteristics. The fused representation is defined in Equation (1).

$$\alpha = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot [f_{\text{conv}} \parallel f_{\text{swin}}]))$$

$$z = \alpha \odot f_{\text{conv}} + (1 - \alpha) \odot f_{\text{swin}}$$

(1)

Here f_{conv} and f_{swin} represent the 768-dimensional feature vectors extracted from the ConvNeXt and Swin Transformer backbones, respectively, while $[f_{\text{conv}} \parallel f_{\text{swin}}]$ represents their concatenation. The matrices W_1 and W_2 are learnable parameters within the two-layer gating network. ReLU was applied to introduce non-linearity, while the Sigmoid function σ mapped its input to the range $[0, 1]$ which produced learned feature weights. The resulting vector α then controlled the relative contribution of each feature representation, and $\odot$ denotes multiplication. The resulting vector z formed the final fused feature representation which was then passed to the final classification layer (Jiang and Ji, 2022).

Both backbones were initialised using weights pretrained on the ImageNet-22K dataset, which contained over 20,000 categories, and were fine-tuned on the ImageNet-1K subset database (Deng *et al.*, 2009; Russakovsky *et al.*, 2014). The combined features were passed into a three-layer multilayer perceptron (MLP) classification head which applied normalisation, additional activation functions, and dropout to improve generalisation and stabilise training. As the data passed through the classifier, the feature size gradually reduced from 768 to 512 to 256 before reaching the final output layer for class prediction. This model was developed from PyTorch v2.11.0 with CUDA 13.0 support (2.11.0+cu130) and TorchVision v0.26.0 (0.26.0+cu130). All experiments were conducted on an NVIDIA Tesla T4 GPU with CUDA acceleration enabled (Paszke *et al.*, 2019).

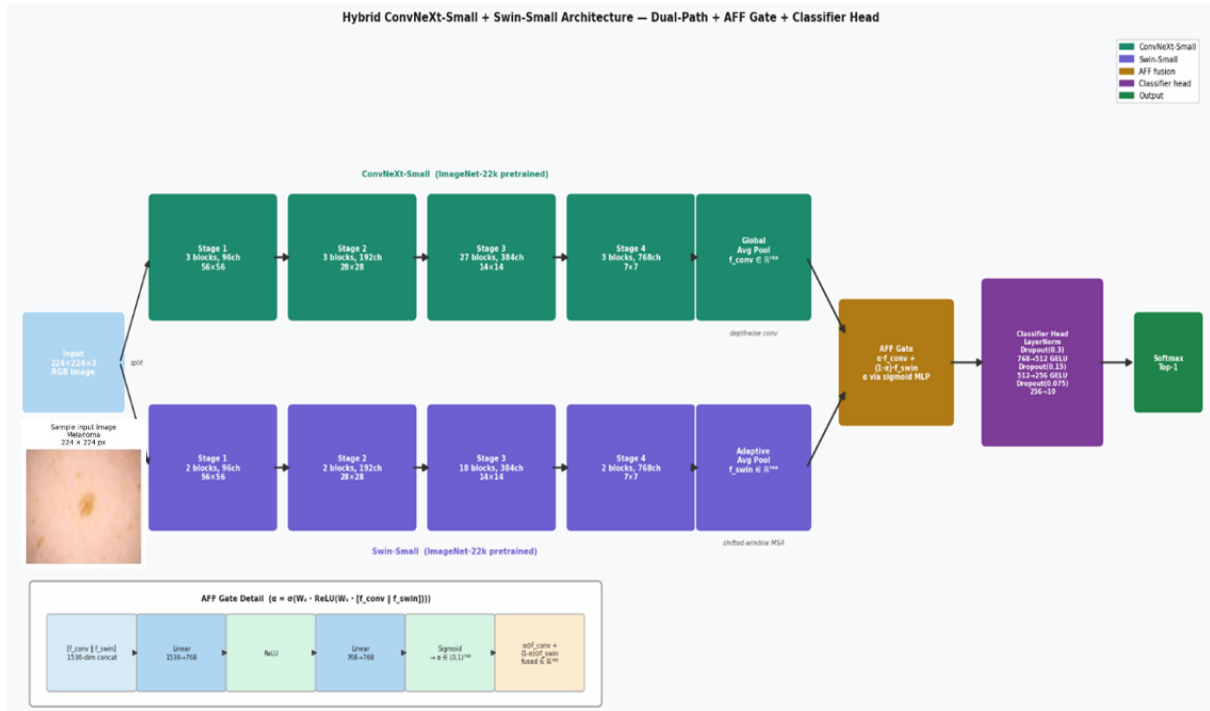


Figure 3.4 Hybrid ConvNeXt+Swin-Small architecture with AFF gate

ResNet-50 Baseline

A ResNet-50 model was selected as the baseline to provide a comparative benchmark upon evaluating the proposed hybrid architecture. To ensure consistency, the model was initialised with pretrained weights from the same ImageNet-1K dataset used for the hybrid model, specifically ResNet50_Weights.IMAGENET1K_V2 (He *et al.*, 2015). The original final fully connected layer was replaced with a dropout layer ($p = 0.3$) followed by a linear classification layer of size $2048 \times \text{num_classes}$ (i.e. number of target classes).

Training was performed in two stages, beginning with a frozen phase to preserve the pretrained features, followed by unfreezing and finetuning with differential learning rates. To maintain uniformity between the two benchmarking models, all optimisation parameters, dataset splits and the training process were kept consistent with those applied to the hybrid framework. The inclusion of a baseline model addressed a common limitation in prior studies, where inconsistent experimental conditions hinder reliable comparison.

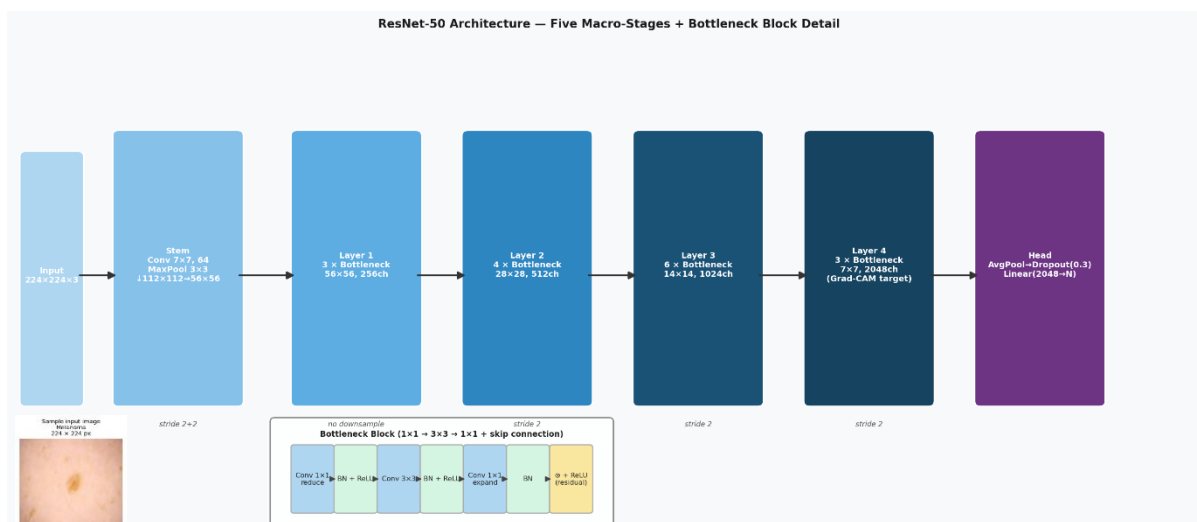


Figure 3.5 ResNet-50 Architecture with bottleneck detail

Table 3.2 Experimental Conditions

Parameter	ResNet-50 Baseline	Hybrid (ConvNeXt+Swin)
Architecture	Single-backbone CNN	Dual-path CNN + Transformer
Backbone	ResNet-50	ConvNeXt-Small + Swin-Small
Pretrained weights	ImageNet-1k V2	ImageNet-22K (if available, else ImageNet-1K)
Input Resolution	224 × 224	224 × 224
Preprocessing	Train: Resize 256 × 256 → RandomCrop 224 × 224; Eval: Resize 224 × 224	Train: Resize 256 × 256 → RandomCrop 224 × 224; Eval: Resize 224 × 224
Feature Representation	Single feature vector (2048)	Fused feature vector (768, via AFF gate)

Parameter	ResNet-50 Baseline	Hybrid (ConvNeXt+Swin)
Fusion Mechanism	None	Attentional Feature Fusion (AFF)
Classification Head	Dropout (0.3) + Linear (2048 → 10)	LayerNorm + MLP (768 → 512 → 256 → 10) with staged Dropout

Table 3.3 Shared Training Hyperparameters

Parameter	Value	Notes
Batch size	32	WeightedRandomSampler for class balance
Max epochs	51	ResNet-50 capped at 51 to match hybrid training length for fair comparison
Warmup epochs (Phase 1)	5	Backbone frozen; AFF gate and head only
Phase 1 learning rate	1×10^{-3}	AdamW; weight decay 1×10^{-2}
Phase 2 learning rates	5×10^{-6} (backbone) · 5×10^{-5} (head/AFF)	$10 \times$ differential; weight decay 5×10^{-3}
Learning Rate scheduler	CosineAnnealingLR	$\eta_{\min} = 1 \times 10^{-7}$
Gradient clipping	max norm = 1.0	Applied per update step
Early stopping	patience = 15, min_delta = 5×10^{-4}	Neither model triggered early stopping; both completed 51 epochs

Parameter	Value	Notes
Loss function	Weighted CrossEntropyLoss	Label smoothing $\epsilon = 0.05$
Checkpoint ensemble	Top 3 · 5-pass TTA	15 forward passes total (3 checkpoints × 5 TTA)

3.4 Training Strategy

Training Strategy and Optimisation

Training was conducted in two distinct phases to preserve pretrained representations. In the initial phase (lasting five epochs), both ConvNeXt-Small and Swin Transformer-Small backbones were frozen. The AFF gate and classification head were the only components to be trained using the AdamW optimiser, employed to update model weights and reduce overfitting (Loshchilov and Hutter, 2017), with a learning rate of 1×10^{-3} and weight decay of 1×10^{-2} . This approach allowed recently added layers to learn task-specific patterns from the dataset, while keeping the pretrained features unchanged.

In the second phase, all layers were unfrozen and fine-tuned with varying learning rates. A higher learning rate of 5×10^{-5} was applied to the AFF gate and classification head, while the learning rate of 5×10^{-6} was applied to the backbone networks. The higher learning rate accelerated learning in task-specific layers, while the lower rate preserved stability in pretrained features. Furthermore, a learning rate scheduling technique was employed in the fine-tuning process, which gradually reduced the learning rate to a minimum of 1×10^{-7} over the course of training.

To further stabilise the training, gradient clipping was applied with a maximum norm of 1.0 to prevent excessively large parameter updates from destabilising the optimisation process. Moreover, dropout was applied progressively across the classification head (at rates of 0.3, 0.15, and 0.075) to discourage the model from relying too heavily on a single feature.

Implementation Decisions and Corrections

Figure 3.6 illustrates the implementation of the proposed ConvNeXt–Swin hybrid architecture. As noted, a dropout rate of 0.4 was reduced to 0.3 to mitigate over-regularisation observed during early training. This implementation directly reflects the architectural and optimisation decisions outlined in Section 3.3 and 3.4.

```
class ConvNeXtSwinHybrid(nn.Module):  
    """  
    Dual-path hybrid: ConvNeXt-Small + Swin-Small fused via AFF.  
    dropout=0.3 (reduced from 0.4 – was over-regularising).  
    feat_dim=768 – matched output dims; projection layers bridge if needed.  
    """  
    def __init__(self, num_classes, dropout=0.3, feat_dim=768):  
        super().__init__()
```

Figure 3.6 Adjustment of dropout rate from 0.4 to 0.3 during implementation refinement

A practical challenge arose from the employment of the Google Colab platform for model training. Training sessions would occasionally be interrupted as runtime sessions would terminate unexpectedly, clearing in-memory variables while preserved model checkpoints on the cloud storage. In the final session of this study, the runtime expired before results could be extracted which led to the loss of the runtime state.

To address this setback, a patch cell (Figure 3.7) was introduced to reconstruct the training state from saved checkpoints using known final values and a lightweight proxy object, that allowed the evaluation pipeline to run unchanged. Moreover, as both the baseline and hybrid model had identical training conditions, the number of training cycles (i.e. epochs) was capped at 51, as shown in Figure 3.8.

```
# — Patch cell – run after history dict above  
BEST_EPOCH      = 48  
BEST_VAL_ACC    = 0.8912  
stopped_epoch   = 51  
MAX_EPOCHS      = 100  
WARMUP_EPOCHS   = 5
```

Figure 3.7 Summary of the training outcomes

```

# Two-Phase Training Setup
# MAX_EPOCHS=51 to match hybrid model training length for fair comparison
MAX_EPOCHS      = 51
WARMUP_EPOCHS   = 5

```

Figure 3.8 Screenshot of the total number of training cycles for ResNet-50 baseline

Class Diagram

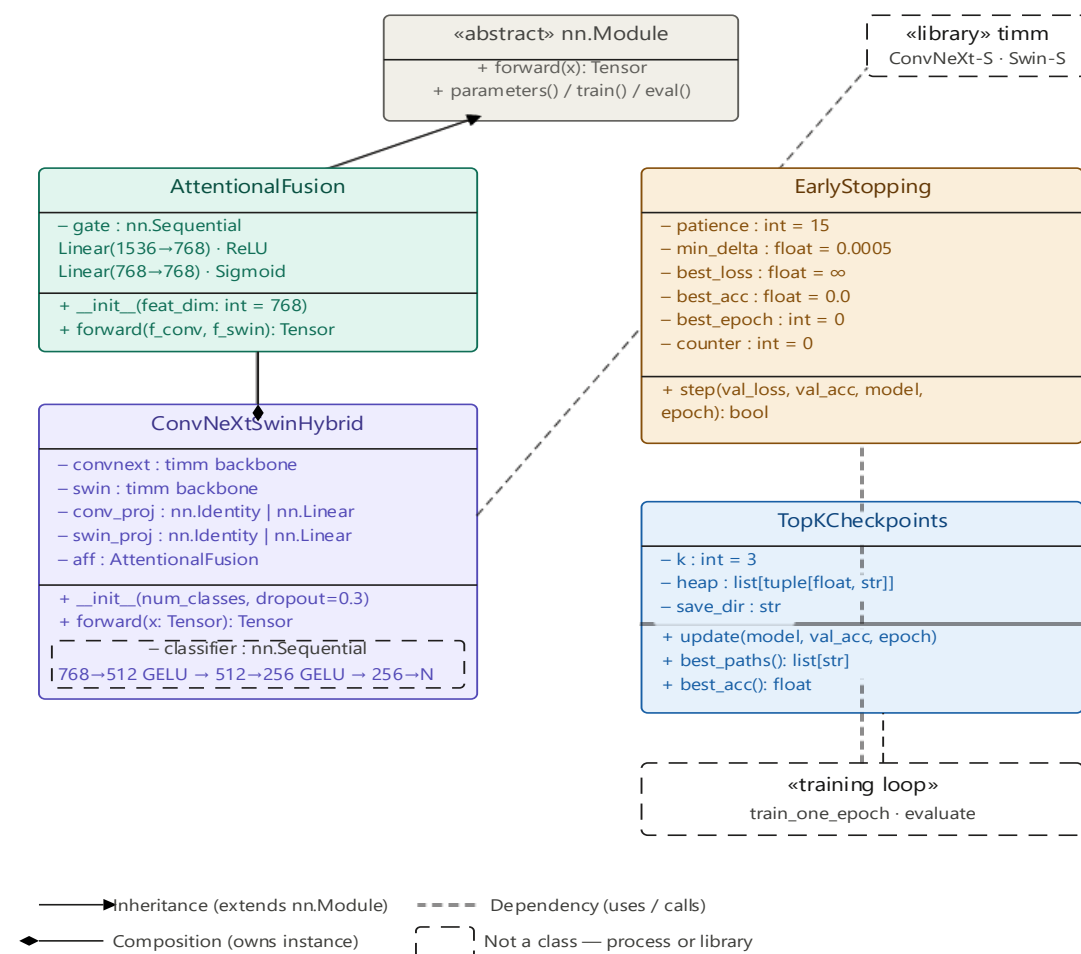


Figure 3-9 Class diagram of the proposed hybrid model architecture, illustrating the relationships between the hybrid network components, feature fusion mechanism, and supporting modules

Class Imbalance Handling

To overcome the challenge posed by class imbalance in the data, two approaches were applied during training. Firstly, a WeightedRandomSampler assisted in balancing the

distribution of classes within each batch, giving greater weight to the classes that appear less frequently during the training process. Following this, the weighted cross-entropy loss function was applied, to place heavy penalties on classification errors associated with majority class samples compared to minority class samples. Moreover, a label smoothing technique was introduced to reduce overfitting and prevent the model from becoming overconfident in its predictions. Collectively, these strategies improved the model's generalisability by promoting balanced learning.

Data Augmentation

All images were resized to 256 pixels before being randomly cropped to 224 pixels to introduce spatial variation during training. Augmentations involved basic transformations, such as horizontal flips, minor rotations up to the degree of 15, as well as minor scaling. In addition, adjustments to saturation, hue, brightening, and contrast were applied to stimulate realistic variations of colour in accordance with real life differences in pigmentation and redness of the lesions.

3.5 Evaluation Framework

Evaluation Metrics

The model's performance was evaluated by involving several metrics: accuracy, macro-average F1-scores, and the area under the receiver operating characteristic curve (AUROC). Although accuracy provides an overall measure of performance and is widely reported in contemporary studies, it can be misleading in imbalanced datasets. This is because high performance on majority classes may obscure poor classification of minority classes, which is relevant for skin disease classification. To mitigate such concerns, the macro-averaged F1 score was included to assign equal importance to each class. This ensured that performance on less frequent conditions, such as Atopic Dermatitis, was not overshadowed by results from more frequent classes, such as Melanocytic Nevi.

Expected Calibration Error (ECE) was introduced to estimate the accuracy and calibration of the predicted probabilities. For example, predictions with 80% confidence should be correct approximately 80% of the time. The correlation between accuracy and confidence increases with decreasing ECE, and values below 0.05 indicate effective calibration (Pavlovic, 2025).

This metric is clinically relevant because commonly reported metrics such as accuracy and AUROC do not indicate whether predicted probabilities are trustworthy enough to guide clinical decision-making in the real world.

Ensemble Strategy

To improve the robustness of the network, an ensemble inference strategy was employed. This approach aggregated the normalised outputs from the top three model checkpoints, selected based on validation performance. In addition, test-time augmentation, previously introduced during the inference stage, was applied and predictions from each augmented version of an input image were averaged to produce the final output. This approach enabled the model to be less affected by small changes in the input, resulting in more consistent predictions.

Skin Tone Stratified Evaluation

To assess model fairness across different skin tones, a stratified evaluation was conducted using the Individual Typology Angle (ITA) as a proxy for skin tone estimation. ITA enabled each image to be quantitatively characterised by a colour measurement, reflecting how light or dark the skin tones were as well as their colour hues. Subsequently, each image was classified into one of the five classes: very light, light, intermediate, tan, and dark. This stratification facilitated a structured comparison of model performance across different skin tone pigmentations in the absence of metadata. The model's performance was subsequently evaluated within each group across metrics such as accuracy and class-wise performance.

Explainability (Grad-CAM)

Model interpretability was assessed by Gradient-weighted Class Activation Mapping (Grad-CAM), which provided visual explanations by highlighting regions of an input image that contributed to most the model's predictions (see Figure 3.10) (Suara *et al.*, 2023). Grad-CAM was executed on the last residual block in the fourth layer of ResNet-50. For the hybrid model, this occurred at the final convolutional layer of ConvNeXt's backbone before global average pooling, since this results in the most semantically relevant visualisations. Accordingly, Grad-CAM offered a qualitative assessment of whether model predictions are based on clinically relevant regions, which produced a greater understanding of its behaviour and transparency.

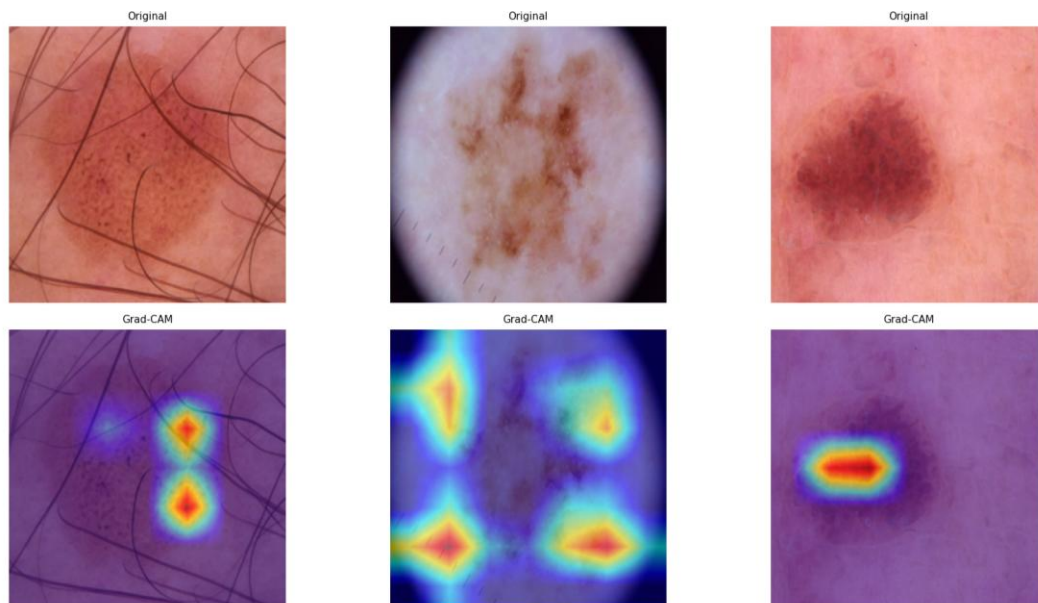


Figure 3.10 Examples of misclassified skin lesions alongside their Grad-CAM visualisations, which highlights the regions the model focused on when making predictions

3.6 Implementation Details

FastAPI Deployment

To demonstrate the hybrid model's practical applicability, a prototype web application was developed by implementing FastAPI (a Python web framework intended for creating web apps) (Siddheshwar Tewari, 2025). FastAPI was selected for its speed and compatibility with machine learning workflows.

The web application consists of a simple frontend interface for uploading images and a backend API that processes the input and performs inference leveraging the trained ConvNeXt–Swin hybrid model. For each input image, the system returns the projected class, risk level, confidence score, as well as clinical recommendation. Moreover, the interface contains a medical disclaimer that makes it clear that the system is not a formal diagnosis and is simply meant for clinical decision support.

It is important to note that a subset of the unseen test dataset was selected when trialling the platform to ensure that predictions were generated from novel data and to verify the system's functionality. Despite the use of ensemble learning and TTA in the evaluation process to maximise performance, the application relies on a single optimised model checkpoint for computational efficiency. This design, therefore, reflects a practical trade-off

widely recognised in publicly deployed applications. A demonstration of the application can be viewed in the attached video file.

The main purpose of this study was to assess the model's performance, and, therefore, developing a fully deployable system was not a priority, given the allocated timeframe for this research. Consequently, factors such as cloud deployment, security, and scalability were deemed outside the purview of this study.

During the design and development of the system architectures and web application prototype, the AI coding assistant Claude Code was used to support implementation tasks such as debugging, refining code, and carrying out quality checks. This allowed for the system's development to be validated; however, all recommendations were manually verified. It should be noted that the use of AI assistance was limited to implementation support and did not replace the initial independent design, experimentation, analysis, or interpretation of results.

User Interface Design

The interface was designed to be engaged with ease to attract and retain technical and nontechnical users. Users can upload an image of a skin lesion and receive a classification along with a confidence score in a short time. Moreover, a feedback option and a clinical disclaimer have been included to support an appropriate interpretation of the results.

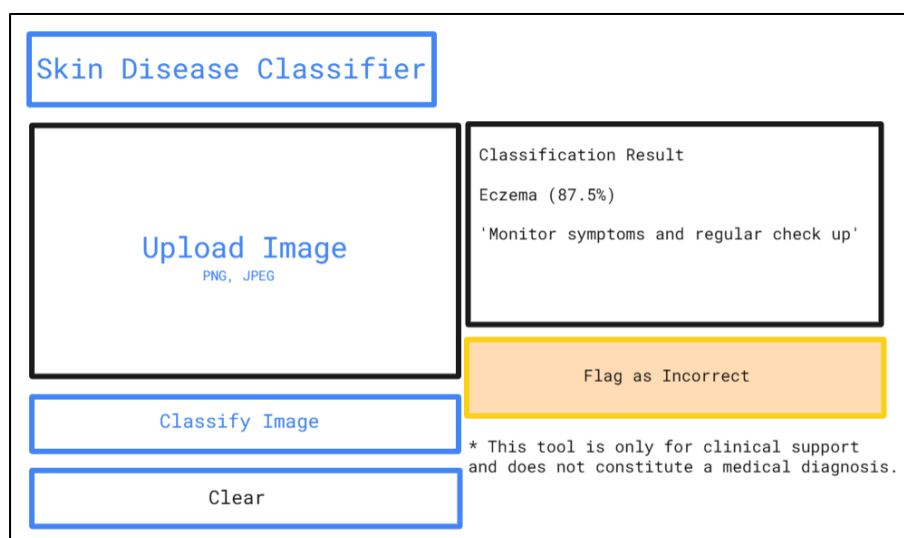


Figure 3.11 Digital sketch of the Skin Disease Classifier interface, showing image upload, classification results, confidence scores, and feedback functionality for misclassified images

Table 3.4 Functional and Non-Functional Requirements

ID	Functional Requirements (What the App Does)	ID	Non-Functional Requirements (How the App Works)
	User actions, classification outputs, risk system, flag feature		Response speed, data privacy, accessibility, safety, system structure
FR1	The system should allow users to upload an image in JPEG, PNG, or WebP format via drag and drop area or file selector	NFR1	The system should populate classification results within 10 seconds of image submission on standard CPU hardware
FR2	The system should classify images into one of ten skin disease categories	NFR2	The system should enable users to upload images up to 10MB in size and reject files exceeding this limit with an error message
FR3	The system should display the top predicted disease class and its confidence score as a percentage	NFR3	The system should not store uploaded images after the prediction has been provided
FR4	The system should assign a risk level based on predicted condition severity	NFR4	The system should run on all modern web browsers (e.g. Chrome or Safari) without requiring additional software installation
FR5	An appropriate clinical recommendation should be displayed based on predicted risk level (e.g. urgent referral for HIGH risk)	NFR5	A disclaimer should appear with every prediction stating that the tool is never to be used as a substitute for medical diagnosis
FR6	The system should display the top five predicted classes with colour-coded confidence bars	NFR6	The interface should support desktop, tablet, and mobile layouts
FR7	The system should enable users to clear existing image and upload a new one without reloading the page	NFR7	The system should be structured into separate files for routing (main.py), inference logic (inference.py), and frontend (index.html) for easier maintenance
FR8	The system should allow users to flag incorrect predictions via a button	NFR8	The system should handle invalid file types by returning an informative error message without crashing

User Stories

The user interaction requirements have been identified based on an analysis of similar dermatology applications including SkinVision, FirstDerm, and NHS111. These platforms focus on providing easy interactions, and results, and offer direct follow-up instructions.

Therefore, the system was designed to deliver predictions that can be easily interpreted, while also offering transparency over uncertain predictions. The overall objective is to support users in making informed decisions, ensuring that the system remains practical, trustworthy, and aligned with real-world expectations and clinical needs.

Table 3.5 User Story Outline

ID	User Story
US1	As a user, I want to upload a skin lesion image and receive a potential diagnosis.
US2	As a user, I want to view confidence scores so I can understand how certain the score is.
US3	As a user, I want to see multiple predicted conditions to consider alternative predictions.
US4	As a user, I want to be notified of high-risk conditions, so I know when to seek urgent medical attention.
US5	As a user, I want to report incorrect predictions so the system can be improved.

3.7 Project Management

Due to the scope of the project, a range of scheduling and task management tools were utilised. For instance, a Gantt Chart was created using Microsoft Excel to schedule tasks, such as dataset preparation, within an allotted timeframe. In addition, Notion AI workspace further assisted in analysing and summarising online reports during the literature review stage.

By adopting an Agile inspired approach, minor tasks were developed iteratively. Lastly, model development was conducted using Google Colab (notebook) with Python, which allowed several attempts of long-lasting training loops to occur, although this presented some practical challenges as reviewed in section 3.4. Nonetheless, frequent evaluation checkpoints assisted in streamlining programming procedures to guarantee effective execution and project success.

Gantt Chart

The Gantt chart below presents the structured timeline for the skin disease classification project and has been organised into key phases including data preparation, model development, evaluation, and reporting.

Skin Disease Classification Gantt Chart													
Task ID	Project Task	W1 01 Feb	W2 08 Feb	W3 15 Feb	W4 22 Feb	W5 01 Mar	W6 08 Mar	W7 15 Mar	W8 22 Mar	W9 29 Mar	W10 05 Apr	W11 12 Apr	W12 19 Apr
1	Dissertation Planning	◆											
2	Literature Review			◆									
3	Data Preparation				◆								
4	Data Preprocessing					◆							
5	Baseline ResNet-50 Model						◆						
6	Hybrid ConvNeXt-Small + Swin-Small Model								◆				
7	Training & Tuning									◆			
8	Performance Evaluation										◆		
9	Web App Deployment											◆	
10	Writing & Submission												◆

Figure 3.12 Gantt Chart illustrating the project timeline (over a 12-week period) for skin disease classification system

Summary

In conclusion, the methodological approach addressed critical shortcomings identified in literature, including class imbalance, restricted evaluation methods, and the absence of fairness assessments in skin disease classification. Therefore, delivering a robust assessment and implementing the hybrid ConvNeXt–Swin architecture presents an overlooked, yet critical, approach to classification research that sufficiently meets the requirements of clinical practice.

4 Results

4.1 Overall Classification Performance

The ResNet-50 baseline and the proposed ConvNeXt–Swin hybrid model were evaluated on a held-out test set of 2,723 images under identical experimental conditions, to ensure a fair comparison. In doing so, performance differences could be attributed to architectural design rather of experimental discrepancies.

The hybrid model achieved a single model test accuracy of 89.39% which increased to 90.27% with ensemble inference and test-time augmentation (TTA). Furthermore, it achieved a macro-averaged F1 of 0.866, and a macro AUROC of 0.980. In comparison, the ResNet-50 baseline achieved 76.79% single model accuracy, 78.48% ensemble accuracy, a macro F1 of 0.745, and a macro AUROC of 0.977. Overall, the hybrid model outperformed the baseline by 11.79 percentage points (pp) in accuracy and 12.10 pp in macro F1, which demonstrated a substantial performance advantage in classification, which was particularly evident in class-balanced metrics.

The improvement in macro F1 indicates that the hybrid model demonstrated a superior predictive capability compared to the baseline model, across all classes, rather than performance being driven mainly by majority categories. This result suggests improved performance across minority classes. These findings are consistent with the observation that many dermatological studies rely solely on accuracy and fail to adequately assess minority class performance or skin tone differences.

The hybrid model achieved a slightly higher macro AUROC than the baseline model. However, the difference was small (0.003) and may not represent a meaningful practical improvement. Consequently, this reinforces the argument that AUROC alone is insufficient for evaluating classification performance in imbalanced datasets, given that it mainly measures a model's ranking performance. Moreover, AUROC is not equivalent to predictive accuracy as a model with a near-perfect AUROC may generate poorly calibrated probability estimates. For instance, the hybrid model's lower calibration error (ECE = 0.0723) provides additional insight for threshold-based decision-making. However, the absence of calibration metrics for ResNet-50 limits direct comparison in this regard.

In summary, these findings suggest improved classification performance in the context of skin disease. Furthermore, the results evidently support the research objectives of developing a robust evaluation framework and a hybrid architecture that can produce reliable predictions. Detailed metrics are provided in Table A1.

Table 4.1 Overall performance metrics

Metric	ResNet-50 Single	ResNet-50 Ens+TTA	Hybrid Single	Hybrid Ens+TTA
Overall Accuracy	0.7679	0.7848	0.8939	0.9027
Macro-Averaged Accuracy	0.7529	—	0.8663	0.8663
Macro F1-Score	0.7268	0.7453	0.8662	0.8662
Macro AUROC	0.9711	0.9766	0.9769	0.9799
Mean Recall	0.7529	—	0.8663	0.8663
Min Recall	0.6040 (Eczema)	0.6040 (Eczema)	0.7240 (Atopic Dermatitis)	0.7240 (Atopic Dermatitis)
Max Recall	0.9710 (Melanoma)	0.9710 (Melanoma)	1.0000 (Melanoma)	1.0000 (Melanoma)

ECE (Calibration)	—	—	0.0723	0.0723
Best val epoch / total	42 / 51	—	48 / 51	—
Best val accuracy	0.7669	—	0.8912	—

Notes: Calibration metrics were not computed for the baseline model.

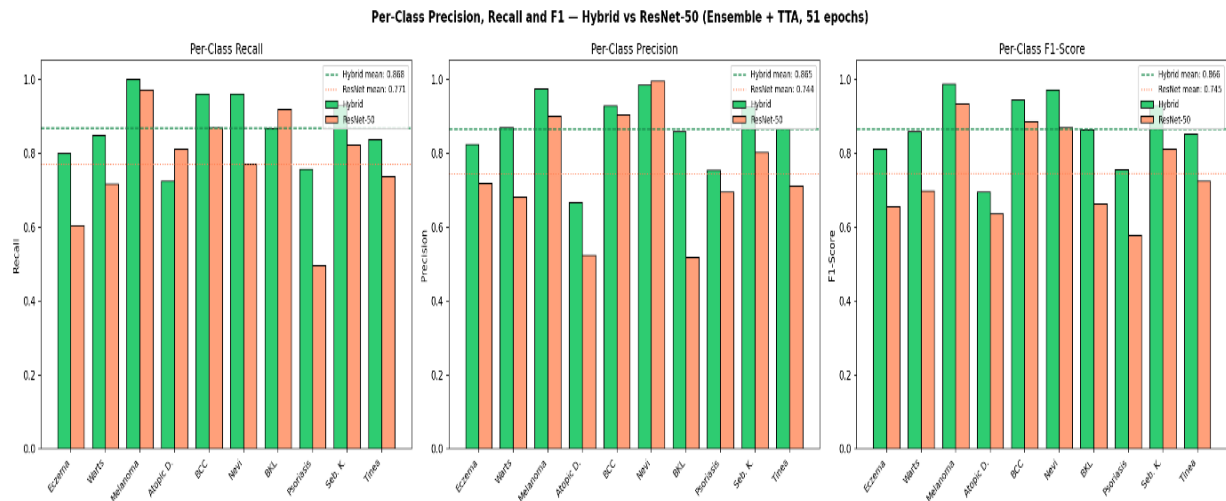


Figure 4.1 Per-Class comparison bar charts between hybrid and baseline model across the metrics recall, precision, and F1 Score

4.2 Baseline Model Performance (ResNet-50)

The ResNet-50 baseline was trained a total of 51 epochs to maintain consistency with the comparative hybrid model. During training, the baseline’s performance remained stable at the latter cycles with the strongest saved checkpoint appearing at the 42nd epoch with a validation accuracy of 0.7669. However, the highest overall validation accuracy was slightly higher at 0.7695 during the 46th epoch. As a result, the best performing checkpoints were identified at the 46th, 42nd, and 40th epoch, with validation accuracies of 0.7695, 0.7669, and 0.7665, respectively. Furthermore, ResNet-50 achieved a test accuracy of 76.79% when applied to a single model, which increased to 78.48% after aggregation and test-time augmentation was applied. It is important to note that neither the baseline nor the hybrid model triggered early stopping, as both completed 51 training cycles.

Despite this, performance varied significantly across classes which resulted in five categories achieving under 80% in recall. This included Psoriasis/Lichen Planus (49.51%), Eczema

(60.36%), Warts / Molluscum (71.56%), Tinea/Candidiasis (73.68%), and Melanocytic Nevi (75.04%). Psoriasis obtained the lowest recall score which could be attributed to its similarity in appearance to other diseases, such as Atopic Dermatitis and Eczema, which are all inflammatory conditions characterised by erythema (i.e. skin redness). This outcome has been addressed in previous studies that have highlighted that convolutional models' dependence on local feature extraction prohibits their ability to capture broader contextual relationships in medical images. From a clinical safety perspective, ResNet-50 also produced high-risk classifications such as predicting Melanocytic Nevi as Melanoma. Notably, such confusions were not observed in the hybrid ConvNeXt-Swin model, which further strengthens the argument that ConvNeXt-Swin provides more reliable separation between visually similar lesion categories.

Despite the reduced results, the ResNet-50 demonstrated competitive performance on certain classes. Markedly, it achieved higher recall scores for Atopic Dermatitis (0.811) and Benign Keratosis (0.919) compared to the hybrid model (0.724). This implies that conventional convolutional models remain effective for conditions that heavily rely on subtle, local texture patterns rather than larger structural patterns across the image. Nonetheless, the overall performance of ResNet-50 illustrates its difficulty in classifying complex and imbalanced classes. Additional baseline fairness trends are shown in Appendix Figure D4.

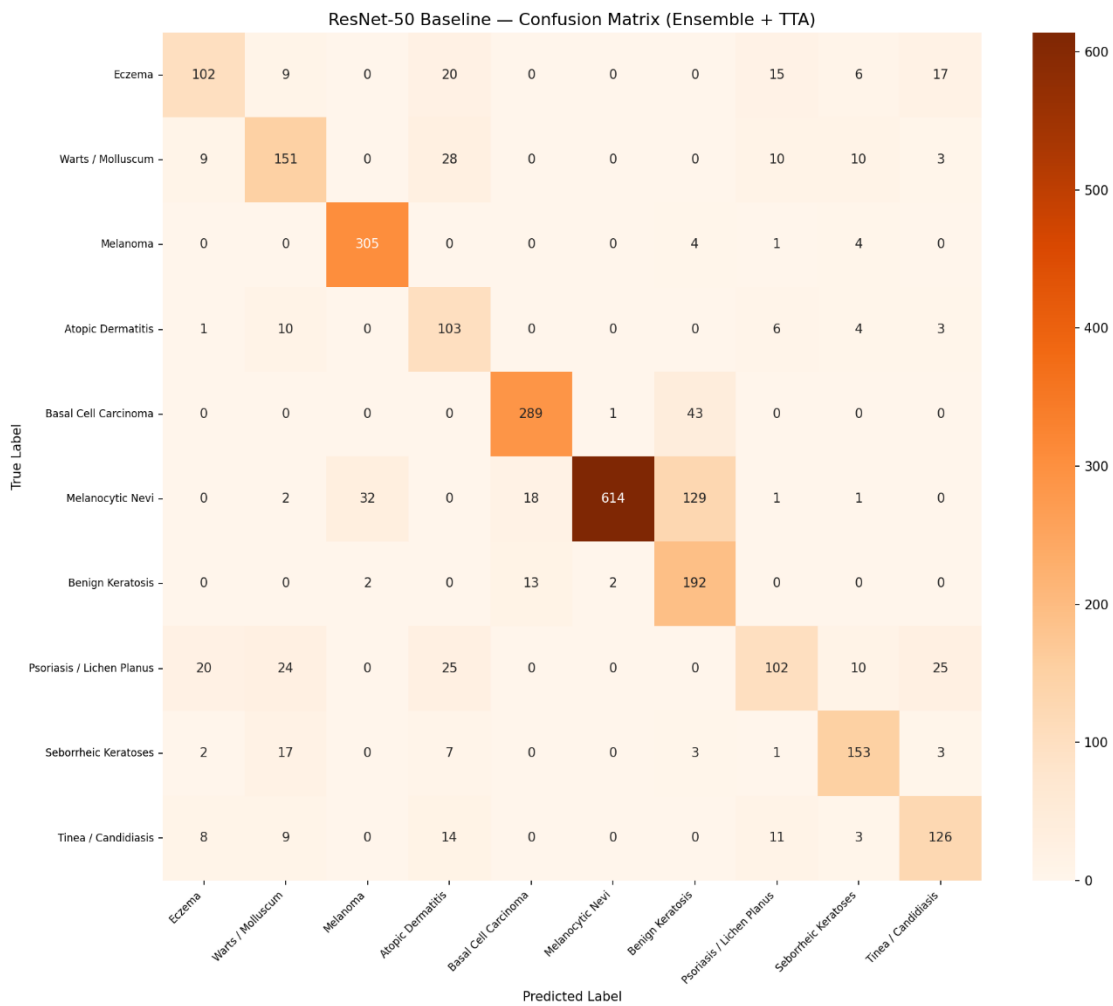


Figure 4.2 ResNet-50 Baseline Confusion Matrix

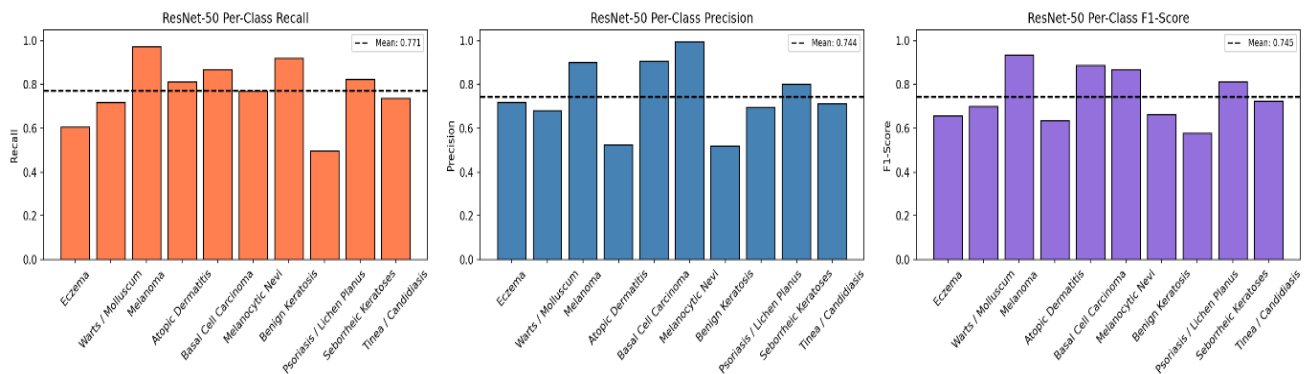


Figure 4.3 ResNet-50 Per-Class performance across recall, precision, and F1 Score

4.3 Per-Class Performance Analysis

Table 7 presents per-class recall for both models across each evaluation setting. These comparisons are based on differences in correct and incorrect predictions for each class, but should be interpreted with caution, as no correction for multiple testing was applied.

Results illustrate that the hybrid model outperformed ResNet-50 in recall across several classes including Eczema, Psoriasis/Lichen Planus, Melanocytic Nevi, Seborrheic Keratoses, Warts/Molluscum, and Tinea/Candidiasis. Overall, the granular analysis reveals that the hybrid model handles complex, context-dependent conditions better, whereas ResNet-50 shows strength on simpler, texture-based lesions. Per-class performance visualisations are provided in Figure D1.

Table 4.2 Per-class recall comparison

Class	ResNet Ens+TTA	Hybrid Ens+TTA
Melanoma	0.9710	1.0000
Basal Cell Carcinoma	0.8680	0.9610
Melanocytic Nevi	0.7704	0.9600
Benign Keratosis (BKL)	0.9190	0.8660
Seborrheic Keratoses	0.8230	0.9300
Warts / Molluscum	0.7160	0.8480
Tinea / Candidiasis	0.7370	0.8360
Atopic Dermatitis	0.8110	0.7240
Psoriasis / Lichen Planus	0.4950	0.7570
Eczema	0.6040	0.7990



Figure 4.4 ConvNeXt-Swin per-class binary confusion matrices

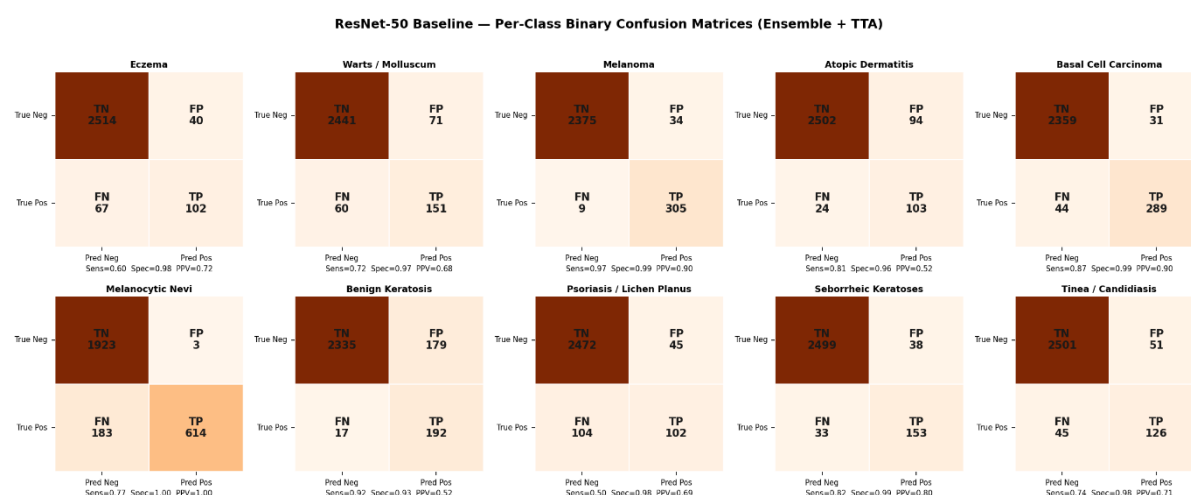


Figure 4.5 ResNet-50 per-class binary confusion matrices

4.4 Attentional Feature Fusion Analysis

Analysis of the Attentional Feature Fusion’s gate offers insight that is not available from the ResNet-50 baseline. Across all ten classes, the average alpha value (i.e. the contribution of the ConvNeXt pathway) consistently remained below 0.5, suggesting a stronger reliance on Swin’s global feature representations. This behaviour was observed during training, as opposed to being predefined, which suggests that the model adaptively prioritises global context.

The gate behaviour provides an explanation for the performance differences across classes. The hybrid model achieved substantial gains in classes, such as Melanocytic Nevi, Psoriasis,

Eczema, and Warts/Molluscum, where overall structure and context is more efficiently captured by Swin’s self-attention mechanism compared to local convolutions. This level of architectural interpretability represents a methodological contribution not seen in the external comparator studies.

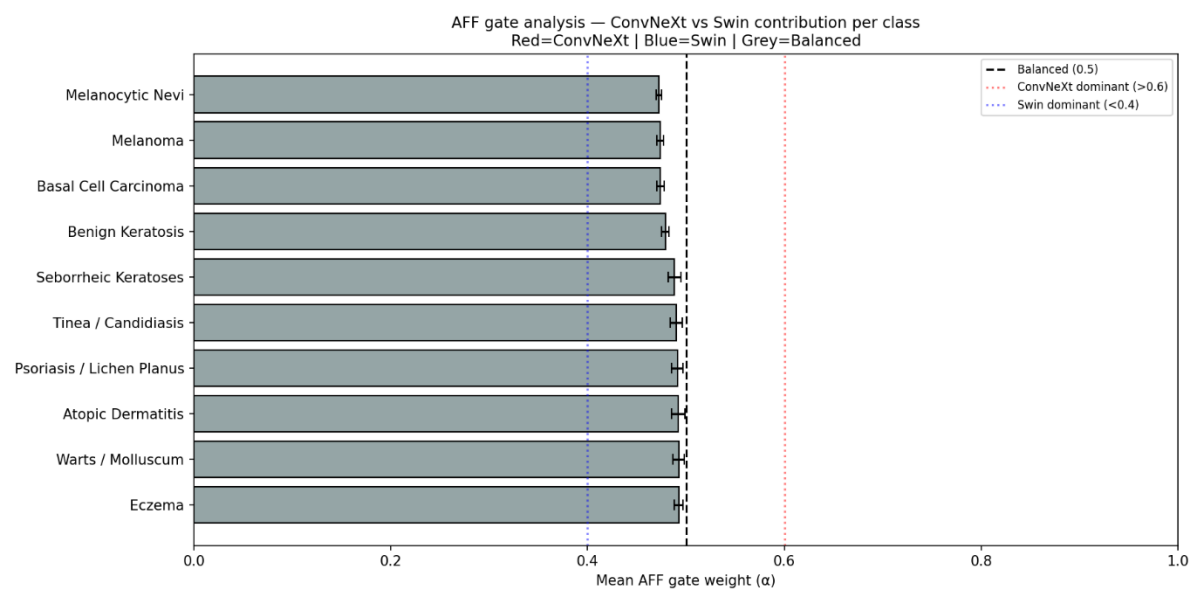


Figure 4.6 Visualisation of AFF gate analysis

4.5 Confidence Calibration Analysis

The Expected Calibration Error (ECE) for the hybrid model was calculated as 0.0723, indicating moderate calibration performance overall. Although values below 0.05 are typically considered well-calibrated, the obtained result suggests that the model’s predicted confidence only partially matched its actual accuracy. The diagram shows partial overconfidence in predictions with lower confidence scores (0.2–0.4), while higher-confidence predictions (0.6–1.0) were better calibrated. In addition, there is a clear separation between correct and incorrect predictions above 0.85 confidence. These results suggest that the model’s confidence scores could still offer clinically useful probability estimates, although further calibration and external validation would be required before real-world clinical use.

Having said that, a drawback in calibration analysis is correlated to the exclusion of this metric for the ResNet-50 baseline. Although a direct comparison cannot be provided,

existing studies have yet to present their models' calibration results, which further strengthens the multi-metric evaluation framework implemented for this research.

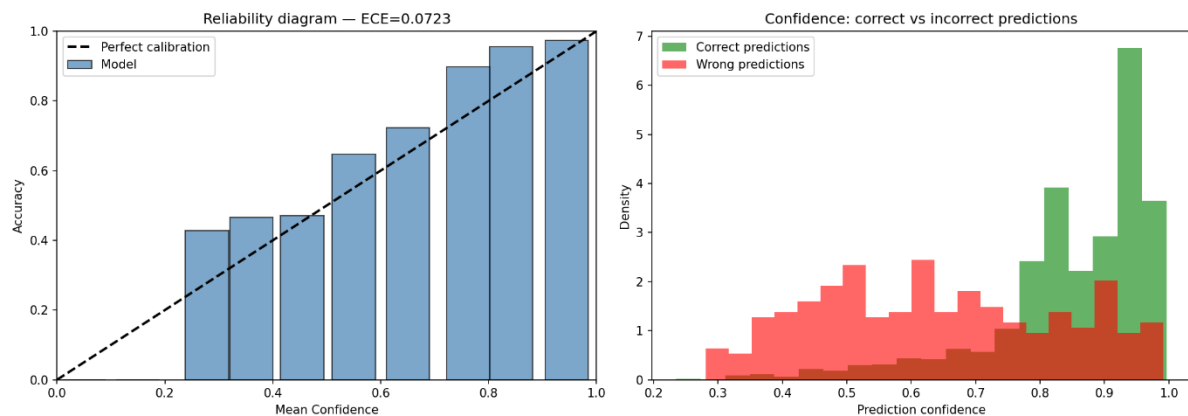


Figure 4.7 Confidence calibration output

4.6 Skin Tone Fairness Analysis

To further support the research objective of developing a robust evaluation framework, both models were assessed across different estimated skin tone groups using the Individual Typology Angle (ITA) method. Based on specific thresholds, images were assigned to one of five groups, involving very light, light, intermediate, tan, and dark. These outputs are reported in Table 8.

Results showed that the hybrid model consistently outperformed the ResNet-50 baseline. In the very light category, ConvNeXt-Swin yielded an accuracy of 90.79%, compared to 80.0% from the baseline. Similarly, in the dark skin tone group, ConvNeXt-Swin reached 88.11% accuracy, whereas ResNet-50 achieved only 72.0%, representing improvements of 10.79 and 16.11 percentage points, respectively. Both models showed reduced performance on darker skin tone groups; however, the decline was smaller for the hybrid architecture (2.68pp vs. 8pp). This demonstrates that the hybrid architecture was less affected by variations across skin tone groups and produced more consistent predictions overall. This pattern supports research stating that architectures that involve global contextual modelling are less sensitive to variations associated with darker skin tones than purely convolutional approaches (Kılıç, 2025). Skin tone analysis visualisations for accuracy and macro F1 results for both models are presented visually in Appendix Figures D2 and D4.

4.7 Grad-CAM Explainability Analysis

Gradient weighted Class Activation Mapping was applied to the most frequently confused class pairs for each model to assess interpretability, and an example can be viewed in Figure 4.8.

For the hybrid model, the main confusion pairs were Atopic Dermatitis misclassified as Psoriasis/Lichen Planus, Benign Keratosis misclassified as Melanocytic Nevi, and Melanocytic Nevi misclassified as Benign Keratosis. In the case of Atopic Dermatitis mistakenly predicted as Psoriasis, activations were randomly distributed across the image, which was likely caused by the model's inability to distinguish between visually similar lesions due to erythema and scaling. In the example of Benign Keratosis misclassified as Melanocytic Nevi, activations were sparse and often focused on hair artefacts and borders of the image. In contrast, Melanocytic Nevi as Benign Keratosis errors showed weak activations, which suggested that the model was uncertain, lacking confidence in its classification.

For the ResNet-50 network, the most frequent confusions were associated with Melanocytic Nevi often mistaken for Benign Keratosis, Melanocytic Nevi misclassified as Melanoma, and Basal Cell Carcinoma misclassified as Benign Keratosis. The activations were found to be more prevalent with an increased spatial concentration. However, in some cases, activation occurred at lesion borders or imaging artefacts. Importantly, the clinically relevant case of Nevi misclassified as Melanoma occurred with high-intensity activations across the lesion body, reflecting genuine morphological similarities.

In summary, the hybrid ConvNeXt-Swin model produced broader and contextually relevant activation patterns, while the ResNet-50 baseline appeared to rely on surrounding regions and image artefacts. This is supported by the differences in misclassifications between Melanocytic Nevi and Benign Keratosis, where the hybrid model produced 17 misclassifications and the baseline model produced 127 misclassifications. This example strengthens the argument of pursuing the combination of local and global feature extraction to ensure clinical relevance, particularly in cases of classifying lesion images.

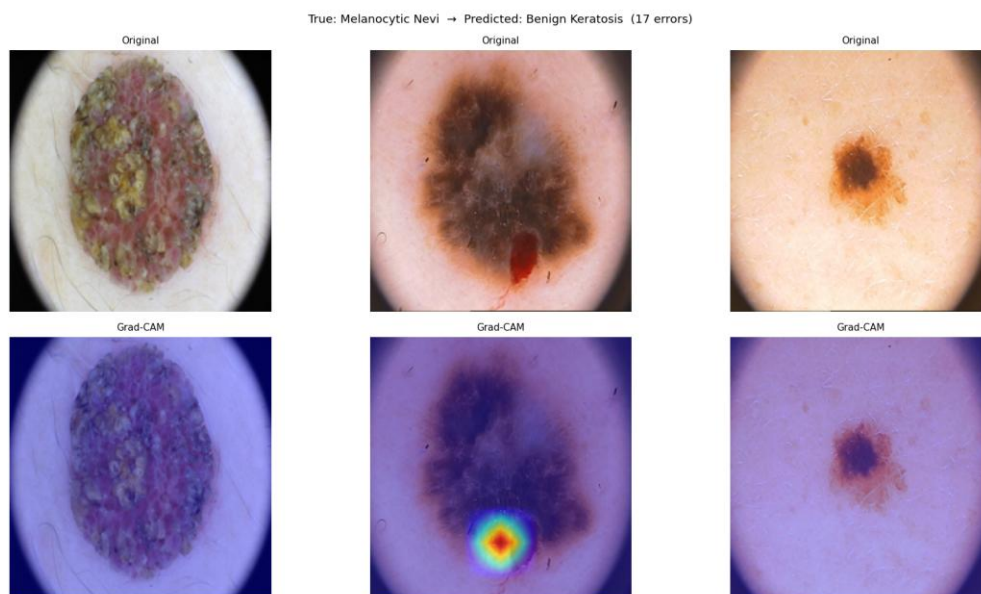


Figure 4.8 Visualisation of Grad-CAM on lesion

4.8 Comparative Analysis with Published Literature

Table 9 compares the results with existing studies that employed the Kaggle Skin Disease data, where the ResNet-50 baseline serves as an internal benchmark. Among the studies included in the comparison table, this dissertation is the sole study to report a wide range of evaluation measures, including macro-averaged F1, ITA-based skin tone analysis, confidence calibration (ECE), per-class sensitivity and specificity, an internal baseline for comparison, and Grad-CAM explainability. As a result, the breadth of this evaluation provides a meaningful methodological contribution to clinically relevant assessment of model reliability, fairness, and interpretability.

Table 4.3 Literature comparison

Study	Architecture	Accuracy	Macro F1	Skin Tone	Cross-Val	Key Limitation
Alruwaili & Mohamed (2025)	EfficientNet ensemble	99.14%	Not reported	None	No	Single split; no macro or fairness analysis

Fernández-Gutiérrez (2025)	ResNet-152	92.30%	Not reported	None	No	2.7 pp train-test gap; no minority analysis
Noor et al. (2025)	CNN + preprocessing	91.64%	Not reported	None	No	70/30 split; aggressive contrast enhancement
This Study — ResNet-50	ResNet-50 (finetuned)	78.48%	0.745	ITA 5-group	No	Single split, internal baseline; identical protocol, early stopping not triggered
This Study — Hybrid ConvNeXt-Swin	ConvNeXt+Swin (AFF)	90.27%	0.866	ITA 5 group	No	Single split evaluation; early stopping not triggered

4.9 Evaluation Limitations

Several limitations should be considered when interpreting these results. Firstly, confidence calibration was not measured for the ResNet-50 baseline, which prevents direct comparison of probability reliability between the two models. Secondly, minor inconsistencies in tone group sample sizes between the two models arose from the ITA proxy being computed independently within each evaluation pipeline. Additionally, to produce a consistent comparison between the two models and ensure that performance differences were solely attributed to architectural design, it was logical and computationally efficient to proceed with a single split. Therefore, neither model was evaluated with cross-validation, which limits the generalisability of the reported metrics. Finally, the ResNet-50 was capped at 51 epochs to match the hybrid model's early stopping point, which may have constrained the

baseline before full convergence and should be considered when interpreting the performance gap.

Summary of Findings

To conclude, the results indicate that the ConvNeXt–Swin hybrid model outperformed ResNet-50, particularly in relation to accuracy and macro F1 score. The performance difference was most evident in complex classes that require an understanding of the entire context. At the per-class level, ResNet-50 achieved higher recall for Atopic Dermatitis (0.811 vs. 0.724) and Benign Keratosis (0.919 vs. 0.866) which suggests a localised advantage for texture-dominant conditions. However, in relation to overall performance, ResNet-50 produced weaker results across imbalanced classes and strict evaluation methods. Nevertheless, several limitations and challenges were identified throughout the evaluation process, giving way to potential improvements and directions for future research which are discussed in the following chapter.

5 Discussion

This chapter critically interprets the findings presented in Chapter 4 within the broader context of skin disease classification and the clinical application of AI in dermatology. The discussion is structured around five distinct themes, covering the architectural factors underlying the hybrid model’s performance gains, the implications of AUROC and macro F1 trade-offs, the practical relevance of the fairness outcomes, the role of calibration and explainability in clinical reliability; and the constraints and future directions of this work. Relevant findings will be linked to methodological design outlined in Chapter 3.

5.1 Architectural Contributions to Performance

The 11.79 percentage point improvement of the hybrid architecture over ResNet-50 cannot be attributed solely to increased model complexity. Rather, it arises from the complementary nature of its feature extraction mechanisms. AFF analysis provided additional insight into model behaviour. Different classes produced different gate weights, which suggested that the network adaptively balances local and global feature representations.

Per-class recall results support this interpretation. The most substantial improvements over the baseline were observed in Psoriasis/Lichen Planus (+26.2 pp), Eczema (+19.5 pp), and

Melanocytic Nevi (+18.9 pp). These are conditions whose diagnosis depends on broader morphological structure and lesion shape rather than localised texture alone. Accordingly, this finding aligns with prior research demonstrating that convolutional architectures are effective at capturing local spatial features (Lecun *et al.*, 1998), while transformer-based models are advantageous at modelling global dependencies within images (Dosovitskiy *et al.*, 2021; Liu *et al.*, 2021). In contrast, ResNet-50 outperformed ConvNeXt in two classes primarily characterised by fine-grained colour and texture, Benign Keratosis (+5.3 pp) and Atopic Dermatitis (+8.7 pp), highlighting the continued advantage of convolutional inductive bias in such scenarios.

This degree of architectural interpretability goes beyond simply reporting better performance. As highlighted in prior research, many dermatological AI studies report improved accuracy without explaining the underlying causes of performance differences, thereby limiting their clinical interpretability. In contrast, the AFF mechanism provides insight into how feature contributions vary across classes, which enables testable predictions about model behaviour.

5.2 AUROC–Macro F1 Trade-off Evaluation

The hybrid model achieved a marginally higher ensemble macro AUROC (0.980) compared to the ResNet-50 baseline (0.977). At face value, this difference is insignificant; statistically, a gap of 0.003 across 2,723 test samples is likely within the expected margin of variation. Nonetheless, without a formal paired test, such as DeLong’s test, this difference cannot be interpreted as a meaningful improvement. In layman’s terms, the difference is likely within the model’s normal uncertainty range, and the confidence intervals for both models would almost certainly overlap. Therefore, this should not be considered strong evidence that the hybrid model is genuinely better on this metric. Furthermore, this highlights another limitation of the study’s evaluation approach, that is the absence of statistical tests to confirm confidence intervals and determine whether correlations are statistically significant. Although other studies have also excluded this level of rigour, including it would serve to strengthen the justification of the proposed model at the institutional level.

From an analytical perspective, AUROC and accuracy capture different aspects of the model’s performance. AUROC evaluates how well a model can separate and rank predictions

across different decision thresholds, yet it fails to provide any insights into the reliability of the prediction confidence score. This is particularly relevant in imbalanced datasets, such as the Kaggle skin disease dataset, where strong results on majority classes can mask poor performance in minority classes, as discussed in Section 2.6. On the contrary, macro F1 is a more suitable evaluation metric because it balances precision and recall while giving equal importance to every class. The hybrid model's considerably higher macro F1 (0.866 vs. 0.745) provides greater insight for this task compared to the relatively small difference in AUROC values. Therefore, it can be concluded that relying solely on AUROC is insufficient when assessing clinical classification models. Instead, class-sensitive metrics and calibration measures provide a more reliable assessment, which directly supports the research objective of developing a clinically applicable diagnostic system. Consequently, macro F1 and class-wise recall provided more meaningful insight into practical classification performance than AUROC alone within this imbalanced dataset.

5.3 Fairness and Bias Mitigation

One of the significant results of this study was the reduction in the disparity of performance between the very light and darker skin tones by the hybrid model with 2.68 pp compared to approximately 8 pp by the baseline model. The result was clinically significant, with the model achieving 88.11% accuracy on the dark skin group by the hybrid model compared to 72.0% by the baseline model.

The increase in fairness gained using the hybrid architecture can be attributed to architectural design. ResNet-50 relies on local convolutional features that may vary with pigmentation (Khalil, Khalil and Ngom, 2023). However, as identified by the AFF gate analysis, the hybrid model leveraged the Swin Transformer pathway for specific classes, allowing the network to focus on shape, boundary, and contextual information that is less sensitive to colour variation (Liu *et al.*, 2021).

A qualitative review of the per-class fairness heatmap suggests that clinically important conditions, such as Melanoma and Basal Cell Carcinoma, show relatively small differences in performance across different skin tone groups. This is clinically relevant, as it indicates that the hybrid model does not disproportionately underperform on high-risk diagnoses in darker-skinned patients. Any remaining disparity in Eczema and Psoriasis cases is expected,

given the known diagnostic difficulty of inflammatory conditions where erythema presentation varies with skin tone (Forsyth *et al.*, 2025).

Having said that, several caveats must be considered given that ITA, used to group skin tones, was computed using the full image rather than a segmented lesion region. As dermoscopic images frequently include background artefacts, shadows, and non-skin areas, these factors may have influenced the colour measurements used in the calculation. Consequently, the estimated ITA values may reflect overall image brightness more strongly than the patient's actual skin tone. This is reflected in the heavily skewed distribution observed, with the very light group comprising 83.7% of the test set (2,279 images) and the dark group 15.1% (412 images), while the remaining groups contained very few samples, making it difficult to draw meaningful statistical conclusions from those categories.

As a result of this limitation, the group assignments contain inherent noise. Consequently, it is not possible to draw definitive conclusions regarding demographic fairness in a clinically meaningful sense. Future work should employ lesion-segmented ITA or clinically validated Fitzpatrick scale labels to enable more reliable evaluation. Despite these limitations, the inclusion of stratified analysis across both models represents a methodological contribution, as comparable studies rarely report demographic performance breakdowns. Supporting visualisations are provided in (Appendix) Figures D2 to D4.

5.4 Clinical Trustworthiness: Calibration and Explainability

Alongside high accuracy, AI-based dermatology tools used in clinical practice also need to provide trustworthy confidence scores and outputs that clinicians can easily understand and interpret. The ECE of the hybrid model at 0.0723 is slightly above the widely accepted threshold of 0.05 for well-calibrated models. However, calibration analysis illustrates that predictions are well-aligned with observed accuracy in the high-confidence range (0.6–1.0), where clinical decisions are most likely to be made. Most incorrect predictions occur in the lower confidence band (0.3–0.6), whereas correct predictions tend to be concentrated between 0.85 and 0.95. This separation suggests that a confidence-based triage system could be implemented, for example by flagging predictions below a threshold (e.g., 0.65) for clinician review.

It is important to note that none of the studies reviewed in the literature review included reliability diagrams or ECE values. This reflects a broader weakness in dermatological AI research, as a highly accurate model that does not report calibration cannot be deployed with confidence at any threshold level, since the relationship between its predicted probabilities and actual accuracy remains unclear. The inclusion of calibration analysis in this study, therefore, directly addresses this research gap and supports the objective of developing a more clinically informative system.

The Grad-CAM analysis reinforces the results due to its interpretability. In relation to the hybrid model, the activation maps for the most confused class pairs target relevant lesion areas, including the lesion body, its boundaries in the case of Nevi misclassified as Benign Keratosis (17 errors), and erythematous patterns in the case of Atopic Dermatitis misclassified as Psoriasis (17 errors). This indicates that diagnosis errors are driven by genuine visual similarities between conditions. Conversely, in the case of ResNet-50 model, activations for high-risk pairs, such as Nevi misclassified as Melanoma (49 errors), often concentrate on peripheral textures rather than the broader lesion context which contributed to more false positives in melanoma diagnosis. The visualisation reveals an inherent weakness as both models demonstrated background activation in some samples, which arose from the absence of explicit lesion segmentation during training. Overall, these results confirm that the hybrid model provides better interpretability and more clinically meaningful decisions, which supports the research objectives.

5.5 Limitations and Future Work

There are several factors that limit the interpretation and generalisation of the results. For instance, the absence of calibration evaluation for the ResNet-50 baseline evidently prevents a direct comparison of probabilistic reliability. Furthermore, although skin tone distribution was consistent between the two models, only the hybrid model received a per-class breakdown per skin tone group. This makes it difficult to conclusively state that the difference in results was due to architecture alone. This reflects a broader challenge identified in the literature, where model performance across different demographic groups is often incomplete or absent.

In relation to the methodology, neither model was subjected to k-fold cross-validation, as both were trained and evaluated on a single 80/10/10 partition. Although early stopping reduces the risk of overfitting, the models have only been tested on one held-out set, which limits the generalisability of the reported metrics. A further limitation relates to the training constraint. The ResNet-50 architecture was limited to 51 epochs, matching that of the hybrid model, although this may not be enough for convergence, and therefore, could have underestimated the baseline results, representing a trade-off between maintaining fairness in the experimental setup and allowing the models to achieve their best possible performance. However, the decision was made deliberately, as training the hybrid model required significant computational power and longer training times.

Considering these limitations, several directions for future work are proposed. Applying post-hoc calibration techniques, such as temperature scaling, would likely reduce ECE potentially below the 0.05 threshold, which would enhance the model's reliability for real-world deployment. Furthermore, including a per-class fairness evaluation for the ResNet-50 baseline would resolve the current evaluation asymmetry. Given the absence of cross-validation, in favourable circumstances, a five-fold cross validation could be considered to yield more statistically reliable performance estimates. Finally, comparing the current AFF gate against alternative fusion approaches (e.g. cross attention) would confirm whether the improved performance is mainly driven by the fusion mechanism itself or by the dual-path architecture. Collectively, these steps would strengthen the validity and clinical applicability of the proposed approach, directly supporting the research objectives.

Summary

The results discussed in Chapter 4 clearly demonstrate that the behaviour of ConvNeXt-Swin is well-explained. The analysis of the Attentional Feature Fusion (AFF) mechanism evidently explained the contribution of both ConvNeXt's and Swin's pathways in producing strong predictive outcomes to visually similar skin diseases. The marginal AUROC difference between models did not alter these conclusions, as AUROC alone cannot fully capture model behaviour in imbalanced clinical settings. In contrast, calibration analysis offered a more meaningful understanding of prediction reliability by showing the importance of assessing model confidence in addition to accuracy.

Furthermore, the smaller performance gap across different skin tones indicated that fairness can be improved through architecture design, rather than relying solely on dataset changes. Although the limitations discussed introduced a degree of uncertainty, they did not change the overall conclusion that the proposed methodology improved the model's robustness, fairness, and reliability.

6 Conclusion

6.1 Revisiting Aims and Objectives

This dissertation set out to build a deep learning framework capable of distinguishing between different types of dermatological diseases, including skin cancer, across diverse skin pigmentations. This was supported by a set of objectives covering model development, evaluation, comparison with a baseline, deployment, and critical analysis.

6.2 Evaluation of Research Objectives

The first objective, reviewing existing research on machine learning for medical image classification, was addressed in Chapter 2. Upon analysing existing literature, several methodological omissions were identified, including data splitting, balanced evaluation, and skin colour assessment. The findings, therefore, informed the methodological and evaluation design which justified proposing a ConvNeXt-Swin hybrid model that leveraged both local and global feature learning.

The second objective was achieved through the design of a hybrid architecture incorporating the Attentional Feature Fusion gate. Subsequently, the third and fourth objectives, robust evaluation and baseline comparison, were addressed by relating performance, fairness, calibration, and explainability metrics to provide a broader assessment beyond overall accuracy alone. Across most evaluation measures, ConvNeXt-Swin outperformed the ResNet-50 baseline, achieving a higher accuracy, macro-F1 score, and reduced performance gaps across skin tones.

The fifth objective, developing a prototype web application, was completed alongside this dissertation. The system provides a functional interface where users can upload dermoscopic images and receive a predicted diagnosis with a confidence score. Finally, the

sixth objective, critically evaluating methodological limitations, is thoroughly discussed in Sections 4.8 and 5.5, where several practical and future recommendations were explored.

6.3 Contribution to the Research Field

This study makes three interconnected contributions. First, from an architectural perspective, the ConvNeXt–Swin hybrid with AFF fusion demonstrated that integrating local convolutional features with global attention mechanisms can enhance performance over a ResNet-50 baseline on a challenging, imbalanced ten-class dataset. This was further supported by the gate analysis, which revealed that the model dynamically balanced its reliance on local and global features depending on the visual characteristics of each lesion class.

The second contribution relates to the evaluation framework. Upon reviewing the publicly available studies containing the same Kaggle dataset, none have reported an equally broad set of metrics, including macro F1, skin tone fairness analysis, calibration (ECE), per-class performance, alongside comparison with an adequate baseline, to this study. Thus, by addressing several evaluation gaps, this dissertation proposes a more inclusive framework that can support future dermatological AI research.

The third contribution is related to clinical relevance. The hybrid model significantly reduces the performance gap across skin tones, narrowing it from approximately 8.0 percentage points (pp) for the ResNet-50 baseline to 2.68 pp for the hybrid when comparing the very light and dark tone groups. This addresses a critical concern surrounding the application of artificial intelligence in dermatology, where models often perform less reliably on darker skin due to dataset imbalance and underrepresentation. The improvement may be attributed to the hybrid model's stronger emphasis on structural and contextual information rather than colour alone.

6.4 Final Reflections and Future Directions

As discussed in the previous chapter, evaluation gaps were identified and explicitly acknowledged. Firstly, calibration metric and per-class breakdown by skin tone group were not assessed for the ResNet-50 baseline, preventing a direct comparison of confidence reliability between models. In addition, there were minor differences in ITA group

distributions, which introduced slight inconsistencies. Moreover, a more rigorous evaluation of skin pigmentation would be better suited for clinical purposes as it would contain verified labels, instead of approximations, such as Fitzpatrick classifications.

Further limitations relate to the selected dataset, and web deployment procedure. Publicly available clinical datasets typically consist of high-quality images; therefore, the model's performance on lower quality smartphone images remains untested, which may affect the reliability of the fully deployed tool. Finally, the web application prototype has yet to undergo clinical validation, and as such, no claims regarding clinical readiness or deployment can be declared. Overall, these limitations introduce a degree of ambiguity into the findings; however, they also highlight important directions for future improvement.

Upon evaluating existing limitations, several areas remain open for future investigation. Undoubtedly, calibration and fairness analyses must be extended to apply to the baseline model, as well as implementing cross validation strategies. Furthermore, isolating the contribution of the AFF gate would provide a stronger explanation confirming which component is directly responsible for driving performance gains. In relation to longitudinal research, external validation on datasets containing verified skin tone labels or metadata is required before live deployment. More broadly, the field needs to shift from prioritising benchmark accuracy towards prioritising clinical reliability.

6.5 Concluding Remarks

Skin disease classification has significant real-world implications, as misdiagnosis can result in delayed treatment, incompatible prescribed medication, as well as unequal support across patients. This dissertation demonstrated that the combination of convolutional and transformer-based architectures can improve classification performance, interpretability, and consistency across different skin tones under realistic evaluation conditions. Although the proposed model is currently clinically undeployable, the findings point to a clear direction that future studies may build upon to help reduce the global burden of skin disease.

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Appendices

- Appendix A: Overall Metrics
- Appendix B: Confusion Matrix
- Appendix C: Training Details
- Appendix D: Additional Evaluation Visualisations

Appendix A: Overall Metrics

Table A1 Evaluation Metrics Glossary

Metric	Type	Description
Overall Accuracy	Classification	The proportion of all predictions that are correct across the entire dataset, regardless of class.
Macro-Averaged Accuracy	Classification	Per-class accuracy computed independently for each class, then averaged equally.
Macro F1-Score	Classification	F1-score is the harmonic mean of precision and recall. It provides a single score that balances the trade-off between the two which is computed per class, then averaged equally.
Macro AUROC	Ranking	Area under the ROC curve, plots the True Positive Rate (Sensitivity) against the False Positive Rate (1 - Specificity). This output is computed per class, then averaged equally.
Mean Recall	Classification	The ratio of True Positives to the total number of Actual Positives. This is computed per class and then averaged equally.
ECE	Calibration	Expected Calibration Error measures how closely a model's predicted confidence probabilities match true empirical accuracy.

Notes: Evaluation metrics explanation derived from Rainio (Rainio, Teuvo and Klén, 2024).

Appendix B: Confusion Matrix

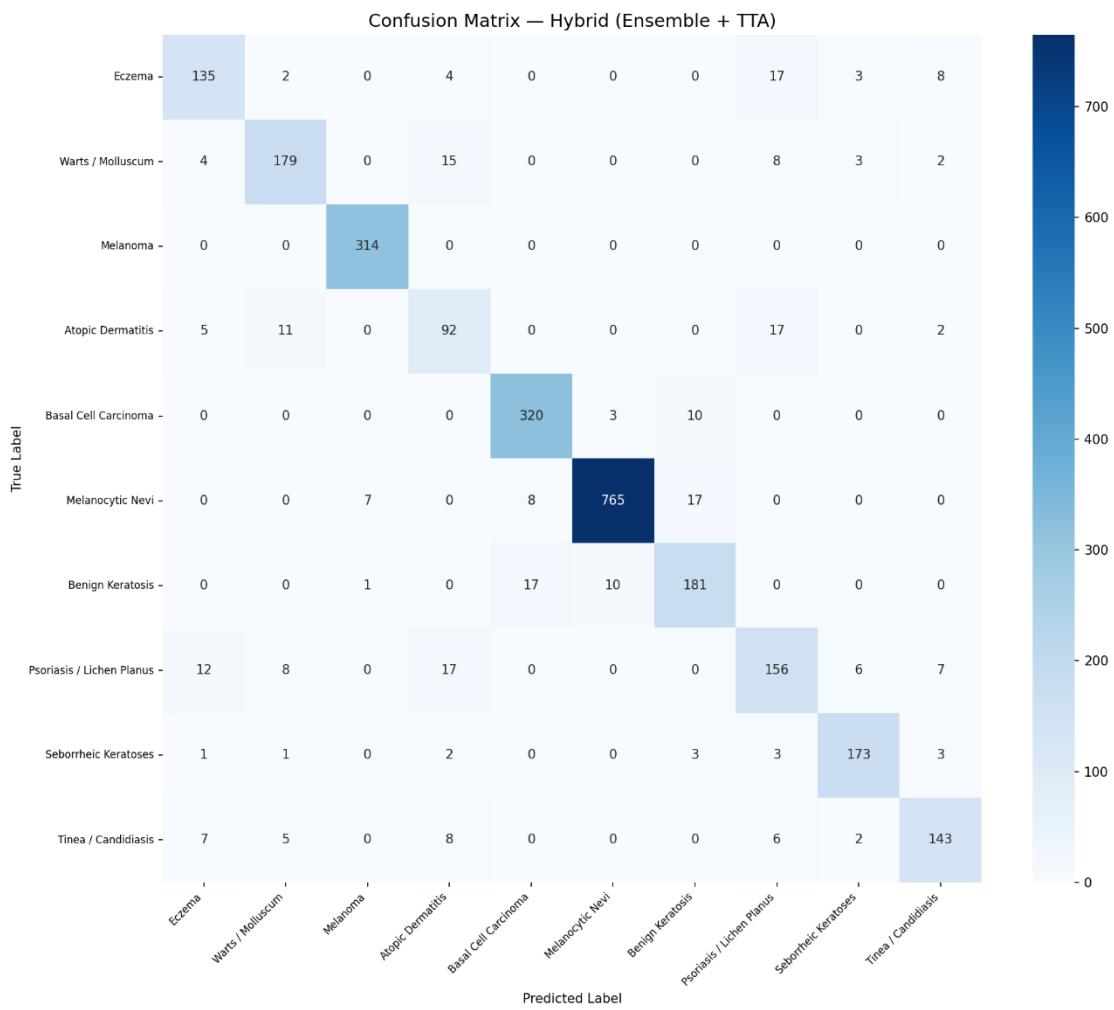


Figure B1 Hybrid ConvNeXt-Swin Confusion Matrix

Appendix C: Training Details

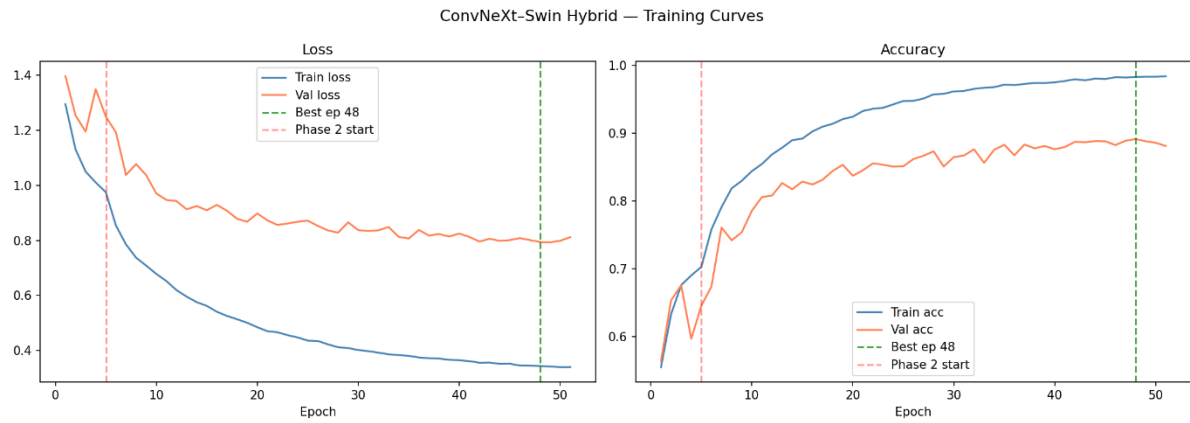


Figure C1 ConvNeXt-Swin Hybrid Training Curves

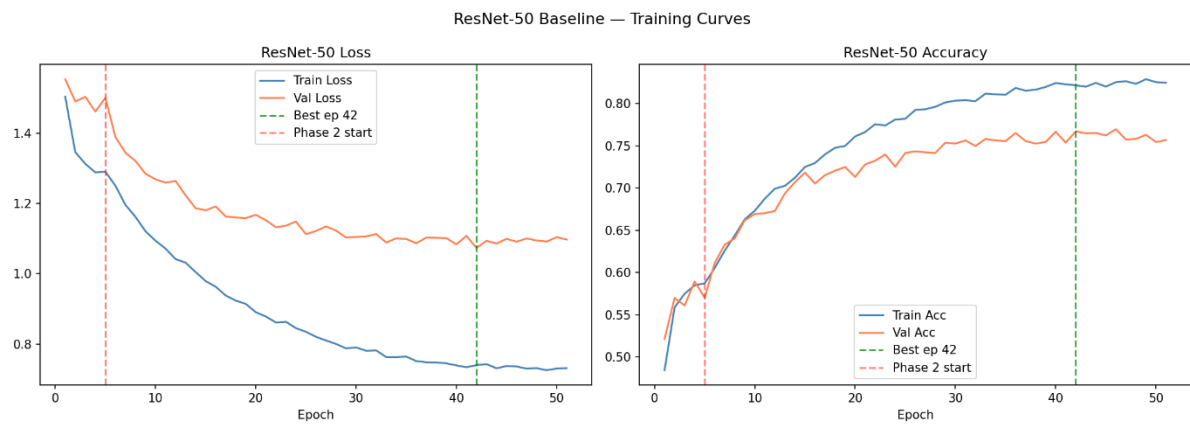


Figure C2 ResNet-50 Baseline Training Curves

Appendix D: Additional Evaluation Visualisations

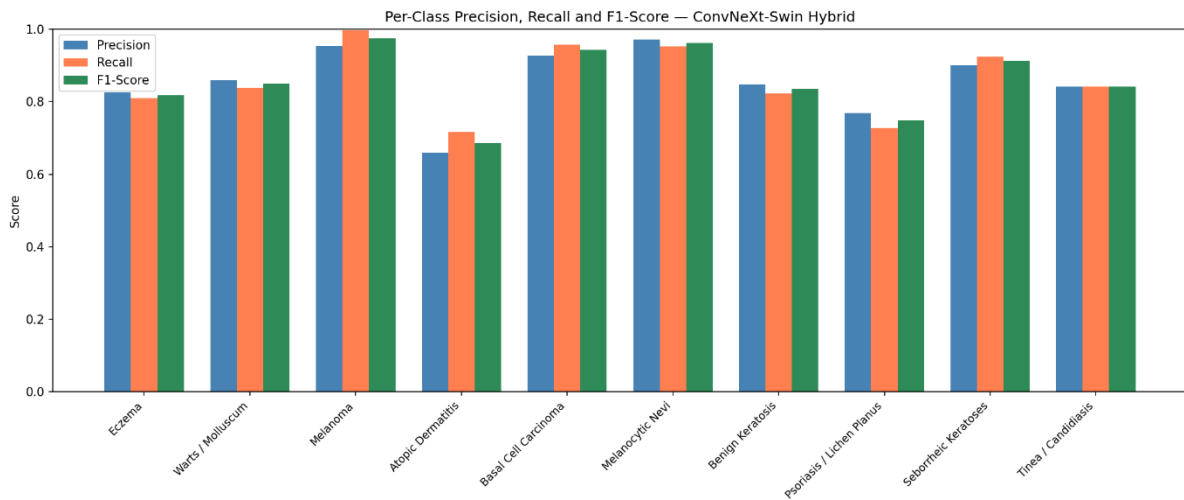


Figure D1 Per-Class Precision, Recall, and F1-Score (ConvNeXt–Swin Hybrid Model)

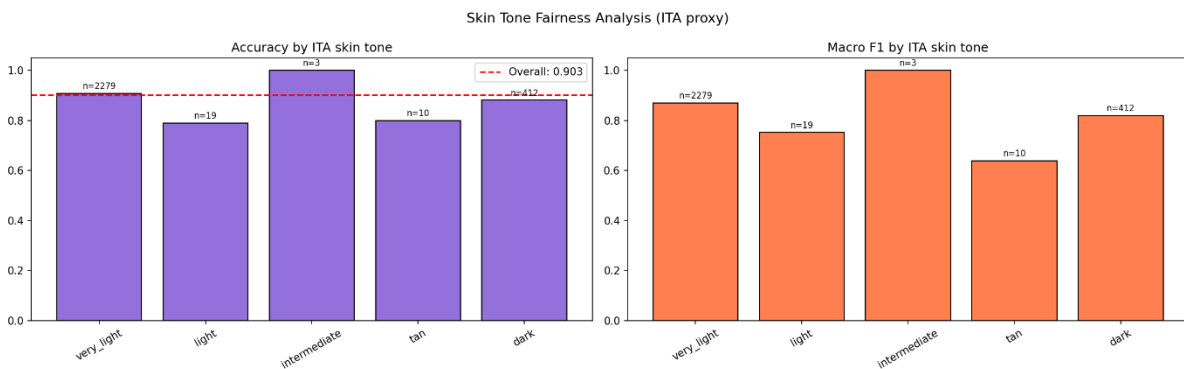


Figure D2 Skin Tone Fairness Analysis (Accuracy and Macro F1 by ITA Group)

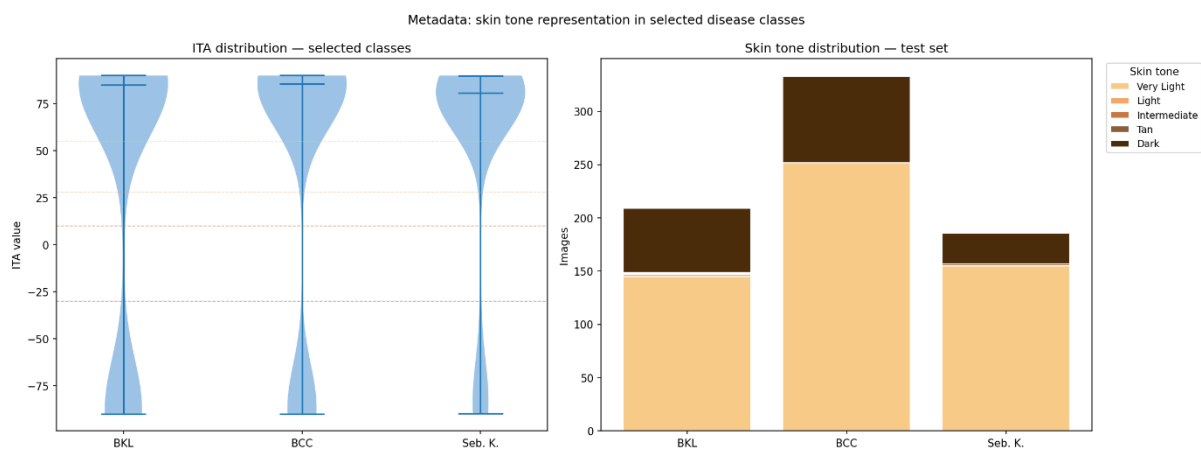


Figure D3 Skin Tone Representation Across Selected Classes

Notes: BKL denotes Benign Keratosis-Like lesions, BCC denotes Basal Cell Carcinoma, and Seb.K. denotes Seborrheic keratosis.

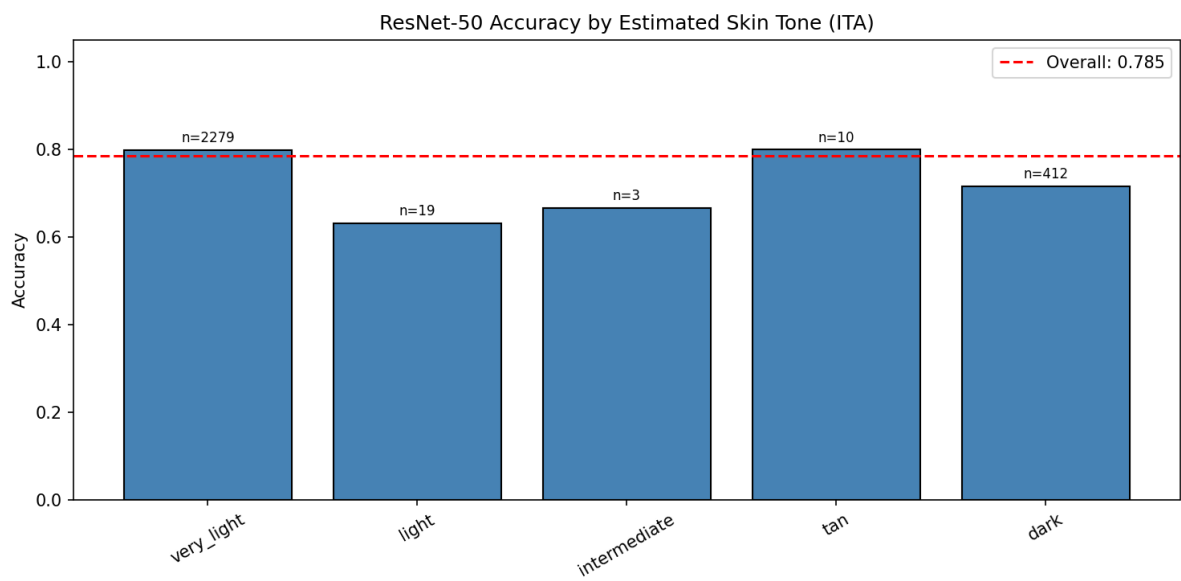


Figure D4 ResNet-50 Accuracy by Estimated Skin Tone (ITA)